

FactoryBench: Evaluating Industrial Machine Understanding

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Code: github.com/Forgis-Labs/Factorybench

Dataset: huggingface.co/datasets/FactoryBench/FactoryBench

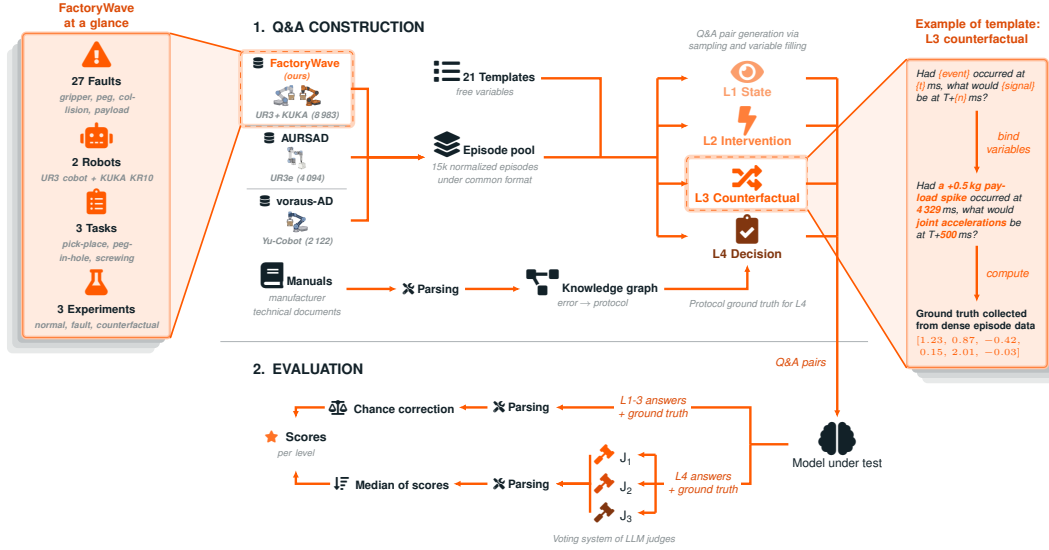


Figure 1: End-to-end FactoryBench pipeline. (1) **Q&A construction**: normalized episodes from FactoryWave, AURSAD [1], and voraus-AD [2] are paired with typed-variable templates (plus the knowledge graph for L4) to generate Q&A items at four reasoning levels. (2) **Evaluation**: the model’s answers are scored deterministically for L1–3 and by the median of three LLM judges for L4 free-form, then chance-corrected per level.

Abstract

We introduce **FactoryBench**, a benchmark for evaluating time-series models and LLMs on machine understanding over industrial robotic telemetry. Q&A pairs are organized along four causal levels (state, intervention, counterfactual, decision) instantiating Pearl’s ladder of causation, and span five answer formats: four structured formats are scored deterministically and free-form answers are scored by an LLM-as-judge voting protocol. We propose a scalable Q&A generation framework built around structured question templates, present **FactoryWave** (a dense, multi-task, multivariate sensor dataset collected from a UR3 cobot and a KUKA KR10 industrial arm), and construct FactoryBench as a large-scale benchmark of over 70k Q&A items grounded in roughly 15k normalized episodes from FactoryWave, AURSAD [1], and voraus-AD [2]. Zero-shot evaluation of six frontier LLMs shows that no model exceeds 50% (chance-corrected) on the structured levels and that the panel ordering reshuffles on the decision-making level, with the L1–L3 leader

falling to the bottom of the L4 ranking; both findings reveal a wide gap between current models and operational machine understanding.

1 Introduction

Modern robotic manufacturing systems emit dense multivariate telemetry, encoding joint states, torques, forces, velocities, contact events, task phases, and fault indicators. Extracting actionable knowledge from these signals is central to monitoring, diagnostics, anomaly detection, and decision support.

Traditional time-series models excel at narrow tasks such as prediction, classification, and anomaly detection [3–5], but they are typically specialized, hard to interpret, output labels without explicit reasoning, and generalize poorly outside their training task [6–8], limiting their utility for automated engineering decisions in a factory setting. Large language models, in contrast, possess strong general reasoning and can produce structured technical explanations [9, 10], but underperform when applied directly to dense numerical time series [11, 12]; specialized transformer-based architectures meanwhile continue to advance time-series forecasting and representation learning [3, 13–19].

Evaluating whether language-based systems truly reason about machine behavior, rather than pattern-matching on textual cues, remains challenging. By *machine-behavior understanding* we mean the ability to (i) interpret the current operational state from raw multivariate signals, (ii) predict the effect of an intervention, (iii) reason counterfactually about alternative histories, and (iv) recommend remedial actions grounded in physical and engineering constraints. How current LLMs perform across these four competences on dense industrial telemetry has not yet been systematically measured. General benchmarks such as MMLU, BIG-bench, and MMMU [20–22] target textual or multimodal understanding and cannot probe machine behavior; closer-in-spirit time-series Q&A benchmarks (TimeSeriesExam [23], ChatTS [24], EngineMT-QA [25], TSAQA [26], TimeMQA [27], MTBench [28], QuAnTS [29]; Table 1) restrict themselves to synthetic or univariate data, omit counterfactual reasoning, or do not involve a robotic system under closed-loop control.

To address this gap we propose FactoryBench, a benchmark for machine-behavior reasoning in time-series models and LLMs. FactoryBench is grounded in **FactoryWave**, a dense multivariate dataset collected from collaborative and industrial one-arm platforms (UR3 at 125 Hz, KUKA KR10 at 83 Hz) with synchronized setpoint, context, feedback, and effort signals. Industrial-robot data of this kind is rare in public benchmarks: such systems live in restricted production environments behind proprietary controllers with limited telemetry and plant-level confidentiality. To our knowledge FactoryWave is the first extensive anomaly dataset to (partly) cover an industrial robot, explicitly bridging the cobot-to-industrial-machine gap. To populate the benchmark, we introduce a scalable question-generation framework of 21 structured templates spanning state, intervention, counterfactual, and decision-making reasoning, applicable to general machine data flows; instantiated over FactoryWave (and AURSAD / voraus-AD), they yield the full FactoryBench Q&A corpus, whose level, answer-format, and context-length distribution we summarise in Appendix B.

2 Related work

Recent advances in LLM reasoning include prompting and deliberative strategies (e.g., Chain-of-Thought and Tree-of-Thought) that improve multi-step performance [9, 10], alongside tool-augmented paradigms such as Toolformer and ReAct that enable grounded computation through external modules [30–32]. These ideas motivate industrial agent design where symbolic reasoning must be combined with numerical workflows.

Time-series modeling has advanced through transformer-based architectures (Informer, Autoformer, FEDformer, PatchTST) [3, 14–16], strong alternative baselines such as TFT and N-BEATS [33, 34], and critical comparisons against simpler linear models [8]. Foundation-model directions (e.g., Chronos) further explore transfer and zero/few-shot adaptation [17, 18], with large-scale industrial time-series corpora such as FactoryNet [35], on which FactoryBench builds. In parallel, anomaly and reliability monitoring literature includes OmniAnomaly, USAD, and Deep SVDD [4, 5, 36], with surveys and benchmarks highlighting persistent gaps in interpretability and root-cause actionability [6, 7].

Causal frameworks provide formal tools for interventions and counterfactuals [37–39], while temporal methods such as Granger causality and modern nonlinear causal discovery support directional reasoning in time series [40, 41]. Knowledge-graph representations further structure domain knowledge for transformer-based reasoning [42]. Existing broad benchmarks (MMLU, BIG-bench, MMMU) [20–22] and classical time-series resources [43–45] do not directly evaluate machine-centered reasoning over industrial telemetry. FactoryBench targets this gap by jointly testing temporal interpretation, causal reasoning, and engineering decision support. Table 1 positions FactoryBench against closely related Q&A and time-series benchmarks along eight axes. The column labels refer to concepts introduced later in the paper: Counterfactual denotes questions that reason about alternative histories (Level 3, Section 3.1); Free-Form denotes open-ended natural-language answers, scored by an LLM-as-judge voting protocol (Appendix A); and Novel Dense TS indicates whether the benchmark contributes a new, high-frequency multivariate dataset rather than relying exclusively on public sources.

Table 1: Comparison of FactoryBench with existing benchmarks. “Counterfactual” = questions reasoning over alternative histories; “Free-Form” = open-ended answers judged by LLMs; “Novel Dense TS” = contributes a new multivariate high-frequency dataset.

Benchmark	Machine/Robot	Multivariate TS	Number of QAs	Counterfactual	Ranking	Numerical	Free-Form	Novel Dense TS
TimeSeriesExam	×	×	~700	×	✓	×	×	×
ChatTS	×	✓	~134k	×	✓	✓	×	×
EngineMT-QA	✓	✓	~110k	×	✓	✓	✓	×
TSAQA	Partial	×	~210k	×	✓	×	×	×
Time-MQA	×	✓	~200k	×	✓	✓	✓	×
MTBench	×	✓	~3k	×	✓	✓	✓	×
QuAnTS	×	✓	~150k	×	✓	✓	✓	×
CLEVR	×	×	~1M	×	×	✓	×	×
CLEVRER	×	×	~305k	✓	×	✓	✓	×
FactoryBench (Ours)	✓	✓	~71k	✓	✓	✓	✓	✓

3 FactoryBench framework

3.1 Four levels of machine understanding

Table 2: FactoryBench levels, what they test, example questions, and expected answer formats.

Level	Type	Example question	What it tests	Answer format
1	State	“We want to isolate the lifting phase in the robot’s time series. Assuming a fixed window length of 15 timesteps, at which timestamp should the window begin?”	Can the model understand what the machine is doing and how it is behaving?	Tensor
2	Intervention	“A collision with a foam cube occurs at $T=850$ ms. Rank signal segments (A–D) in the order you would expect them to appear after the event.”	Can the model reason about how the machine will behave in the face of an event?	Ranking
3	Counterfactual	“Had a payload misconfiguration occurred at $T=200$ ms, what would the target torque on joint 2 at $T+50$ ms have been in this counterfactual case?”	Can the model reason accurately about theoretical scenarios?	Scalar
4	Decision	“Given the sensor stream below, does the machine show signs of anomalous behavior? If yes, identify the root cause and the steps to fix it.”	Can the model make informed decisions about the machine?	Free-form

Our four-tier hierarchy is explicitly built on top of Pearl’s ladder of causation association, intervention, and counterfactual reasoning which has become the organizing principle for causal inference and

is increasingly adopted as an evaluation axis for machine-learning systems [37, 38, 46–48]. We extend the causal hierarchy with a fourth decision-making level that reflects how industrial robotic platforms are actually operated: after interpreting state and causal relationships, operators must select and execute corrective procedures, typically prescribed by vendor manuals (e.g., Universal Robots error-code handbooks). This final level aligns with the diagnose-then-act loop that is standard in fault-tolerant control. Grounding the hierarchy in these established principles ensures that each level probes a qualitatively distinct reasoning skill and that failure modes are interpretable in terms of the underlying causal or decision-theoretic primitive.

To systematically evaluate machine-behavior reasoning, we organize question-answering tasks according to this four-tier hierarchy, each probing distinct reasoning capabilities:

Level 1: State. This level assesses the agent’s ability to interpret the current state of the machine under normal conditions, including identification of operational modes, prediction and recognition of sensor patterns. Questions at this level require accurate extraction and interpretation of time series features.

Level 2: Intervention. At this level, the agent must reason about the consequences of interventions or events occurring at the present timestep that might perturb the distribution of machine states. Tasks include predicting the immediate impact of control actions, diagnosing faults as they arise, and understanding causal relationships in real time for both normal and anomalous episodes.

Level 3: Counterfactual. This level evaluates the agent’s ability to reason about hypothetical scenarios, such as the effect of an event or intervention at a previous timestep. Questions require the agent to simulate alternative histories and assess how outcomes would differ under counterfactual conditions, while still considering the history they know.

Level 4: Decision Making. The highest level evaluates the agent’s ability to act on its understanding of the system. Given a recorded episode and a task description, the agent must produce a sequence of actions or recommendations to answer that prompt. In FactoryBench, this targets troubleshooting and optimization, and combines the skills exercised at the previous three levels: reading the system state, reasoning about causes, and weighing alternative interventions. It reflects what we expect from an LLM acting as an engineering assistant on the factory floor.

By structuring Q&A tasks along these four levels, FactoryBench enables rigorous and granular assessment of machine-behavior reasoning, from basic state recognition to advanced decision support. Figure 6 (Appendix B.1) instantiates Pearl’s three causal levels (L1–L3) on robotic time-series data and adds the L4 decision-making layer that FactoryBench introduces on top. Each question is emitted in one of five answer formats (single-select MCQ, multi-select MCQ, ranking, tensor, free-form); the four structured formats are scored deterministically and free-form answers are scored by an LLM-as-judge voting protocol, with per-format details in Appendix A and an analysis of the three judges’ inter-rater reliability in Appendix A.1.

3.2 Time series data and causal schema (SCE)

So that the dataset structure itself encodes causality, all time-series signals are organized into three causal groups: **Setpoint** (the controller’s commanded target), **Context** (physical and configured conditions, split into static episode metadata and dynamic time-series channels), and **Effort+Feedback** (the machine’s measured response). Under healthy operation the response is a known function of setpoint and context; faults manifest as structured residuals from that baseline, giving a single operational fault definition that applies across machines and tasks. Full per-group channel mappings, the formal residual definition, and per-robot calibration details are deferred to Appendix M.

The data conforms to this schema across two source types:

- **FactoryWave:** A custom dataset generated from one-arm robotic platforms executing industrial tasks (e.g., pick-and-place) under varied conditions and systematically injected anomalies.
- **Open Source Datasets:** Adapted datasets such as AURSAD [1] and voraus-AD [2], offering diverse industrial scenarios and preprocessed to conform to the unified episode structure.

134 The datasets integrated into FactoryBench are summarized in Table 3; all provide the full setpoint-
 135 context-effort triplet at 83 Hz or higher and have been (re)labelled to conform to our unified episode
 136 schema.

Table 3: Datasets incorporated into FactoryBench. FactoryWave is collected and released with this work; the other two are open-source datasets adapted to our schema. Tasks: PnP = pick-and-place, Scr = screwing, PiH = peg-in-hole.

Dataset	Robot	Episodes	Frequency	Tasks	Anomalies
AURSAD [1]	UR3e	4094	100 Hz	Scr	4
voraus-AD [2]	Yu-Cobot	2122	100/500 Hz	PnP	12
FactoryWave (ours)	UR3, KUKA KR10	8983	125/83 Hz	PnP, Scr, PiH	27

137 4 Reliable Q&A generation

138 4.1 Scalable generation via extensive labeling

139 Scalable ground-truth generation is the central challenge of any Q&A benchmark grounded in raw
 140 data. FactoryBench addresses this by coupling a structured labeling ontology with a context-free
 141 grammar (CFG)-style template system, designed by robotics experts and PhD researchers. Each
 142 template contains variable slots filled at generation time from one of two sources: label information
 143 read directly from the episode data (signal names, timestamps, anomaly labels, root causes), or a
 144 predefined sampling distribution attached to that slot (prediction horizons, numerical thresholds,
 145 curated vocabularies for discrete options). The discrete vocabularies are seeded from the label sets of
 146 AURSAD [1] and voraus-AD [2], extended to cover FactoryWave-specific cases, and released in full
 147 so every instantiation is inspectable and reproducible. The context time series used to fill the variables
 148 is sampled uniformly by data source (open source vs FactoryWave), then dataset, experiment, length
 149 within controlled bounds, and finally placement, yielding a combinatorial expansion from a small set
 150 of carefully designed templates into a large, diverse question pool.

151 Each of the 21 templates is manually authored to probe one specific machine-understanding level
 152 while remaining general enough to admit a wide range of instantiations across episode segments,
 153 signal names, event descriptions, timestamps, and predicted values. Answer options for single-
 154 and multi-select questions are drawn from a shared pool of pre-defined verifiable statements (for
 155 example, the question “*What anomaly is present in this robotic sensor time series?*” draws its options
 156 from the full catalogue of documented anomalies), each paired with a rule that can be evaluated
 157 deterministically against the time series given the densely labelled data, so ground truth is never
 158 imputed or inferred but computed directly from the underlying labels; for single-select MCQ items,
 159 the option-sampling step additionally enforces that exactly one of the four sampled options evaluates
 160 to true on the chosen episode, so the question is unambiguous by construction. This is also our
 161 primary defense against the natural concern that template-generated benchmarks can be solved by
 162 learning template surface structure rather than the underlying reasoning: every template admits a
 163 combinatorially large instantiation space over signal names, timestamps, prediction horizons, payload
 164 conditions, and injected-fault types, so two questions sharing the same template rarely overlap in more
 165 than a fraction of their variable slots, and the correct answer is never recoverable from the question
 166 text alone since it is always determined by the paired time series, which varies across instantiations.

167 4.2 Density of FactoryWave

168 FactoryWave is collected from two physical robots (UR3 and the KUKA KR10; Figure 2) executing
 169 up to three industrial tasks each: pick-and-place, screwing, and peg-in-hole. Each complete task
 170 cycle constitutes one episode, recorded at 125 and 83 Hz across over 100 sensor channels covering
 171 joint setpoints, feedback, torques, speeds, estimated contact forces, TCP pose, gripper state, and
 172 task-phase labels. To compensate for the limited tool-side telemetry exposed by the KUKA KSS
 173 controller, we additionally mount a 9-DoF inertial measurement unit on the KUKA gripper and
 174 stream its acceleration, angular-rate, and orientation channels in sync with the controller signals.
 175 The full per-robot, per-task, per-condition breakdown of the 8,983 episodes is given in Table 9 of
 176 Appendix G.



Figure 2: FactoryWave tasks and fault catalogue. Photographs of the robotic platforms (UR3, KUKA KR10) executing the three tasks (pick-and-place, peg-in-hole, screwdriving) and a representative subset of the 27 systematically injected fault conditions.

Episodes are organized into three experiment types. **Normal** episodes capture nominal operation across three payload levels (light, medium, heavy for pick-and-place). **Fault** episodes inject one of 27 fault types (spanning gripper failures, misconfigurations, collisions, and peg-in-hole-specific anomalies), each recorded with fault-specific metadata and, where applicable, a precise injection timestep. **Counterfactual** episodes provide ground truth for Level 3 causal reasoning by approximating the do-operation on physical hardware, via the protocol described next.

Counterfactual generation approximates the last step of Pearl’s ladder on physical hardware, which cannot reproduce a bit-identical pre-injection state. For each baseline we re-execute the task 3–5 times with every controllable condition held fixed (object pose, payload, controller configuration, exact scripted trajectory) and inject the target fault at a fixed timestep t . We then keep the run whose pre-injection segment minimizes the signature kernel MMD against the baseline, and use that episode as an approximate ground truth for hypothetical divergence of baseline.

4.3 Knowledge graph and protocol extraction

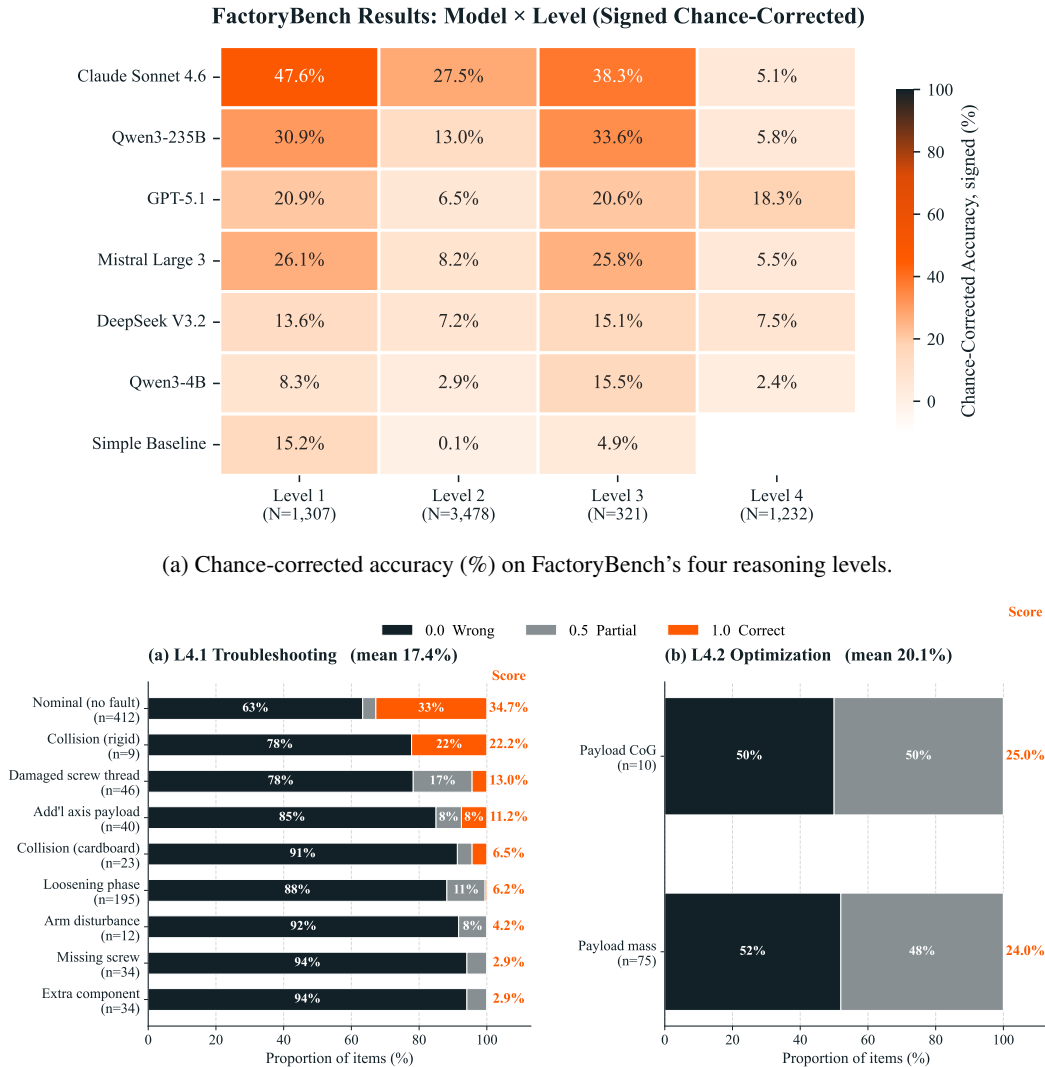
Reliable ground truth for Level 4 troubleshooting questions requires more than episode labels: it demands machine-specific recovery protocols grounded in manufacturer documentation. For each robot in FactoryBench we parse the manufacturer’s official runtime error documentation into a structured mapping from error codes to error names, descriptions, and recommended recovery steps, capturing what the controller reports when a fault occurs and what an operator should do to resolve it. We then map every anomaly type studied in FactoryBench to the most likely corresponding manufacturer error code, with PhD-level robotics experts reasoning about the physical cause of each anomaly and identifying which runtime error it would most plausibly trigger on the target robot. Where a physical fault injection (such as gripper misactivation, peg misalignment, or external collision) has no well-defined controller error, the same experts author the recovery protocol directly using the same structure for consistency.

The complete mapping (error-derived and expert-authored protocols alike) is ingested into a knowledge graph alongside robot specifications, task definitions, and signal schema. At question generation time, the pipeline queries this graph to automatically assemble the relevant protocol context and inject it as ground truth into Level 4 troubleshooting Q&A pairs, without requiring per-question human review.

206 5 Evaluation and results

207 5.1 Zero-shot evaluation

208 We evaluate a cross-vendor panel of frontier LLMs available at submission time (April 2026): Claude
 209 Sonnet 4.6 [49] and GPT-5.1 [50] on the closed-weight side, and DeepSeek V3.2 [51], Mistral Large
 210 3 [52], and Qwen3-235B [53] on the open-weight side. We also include Qwen3-4B [53] in zero-shot
 211 mode as a lightweight open-weight reference point. For calibration we report a non-LLM baseline
 212 that dispatches per item by answer format: linear regression on the target signal for scalar and tensor
 213 templates, and uniform random over the option set for single- and multi-select MCQ and ranking
 214 templates. Free-form Level 4 templates are excluded from the baseline (no traditional method maps
 215 cleanly to producing a remediation protocol). Figure 3 reports chance-corrected accuracy for the full
 216 panel.



(b) GPT-5.1 L4 score distribution by ground-truth root cause (left, troubleshooting) and misconfigured parameter (right, optimization). Each bar shows the proportion of items scored 0/0.5/1; the right-edge value is the per-category mean accuracy.

Figure 3: Main results. (a) Cross-model panel performance across all four reasoning levels. (b) Per-fault and per-parameter breakdown of the GPT-5.1 L4 mass.

217 5.2 Signal comprehension across Levels 1–3

218 **Most models fail to separate from the simple baseline on Level 1, including the largest in the**
219 **panel.** Under the signed chance correction the non-LLM baseline lands at +12.2% on L1 (a value
220 we treat as the empirical zero). Two of six LLMs fail to clear it: Qwen3-4B (+5.9%) and DeepSeek
221 V3.2 (+10.8%) score below the baseline. GPT-5.1 (+19.0%), the largest model in the panel by
222 parameter-count and inference-cost proxies, clears it only modestly; Mistral Large 3 (+24.8%),
223 Qwen3-235B (+29.7%), and Claude Sonnet 4.6 (+46.8%) establish a substantial margin. Raw scale
224 does not reliably confer the ability to extract precise state from dense industrial signals; L1 measures
225 something general-purpose pretraining does not optimize for.

226 **Models that struggle on Level 1 still outperform the baseline on Levels 2 and 3.** DeepSeek V3.2
227 (+4.1%, +12.8%) and GPT-5.1 (+4.0%, +18.2%) both lift above the L2/L3 baseline audit values
228 (−3.0%, +1.7%) despite trailing on L1. That being said, the model rank order is nearly identical
229 across L1–L3: Claude Sonnet 4.6 stays in the lead at L2 and L3 (+25.8%, +35.8%), Qwen3-235B is
230 the strongest open-weight model and reaches +30.9% on L3, and Mistral Large 3 follows (+5.6%,
231 +23.4%). This shows a strong correlation between the score of Levels 1–3, and points towards
232 the fact that a stronger state comprehension translates into better intervention and counterfactual
233 reasoning: L2 and L3 build on L1 ability rather than substituting for it.

234 **A human expert reaches near-ceiling on Levels 1–2, but not on Level 3.** To establish what score
235 is actually achievable on these items, a robotics expert answered a 60-item sample under open-tool,
236 no-LLM conditions, graded by the benchmark’s own scorer (Appendix F.2). On raw scores over
237 the identical items the expert reaches 0.962 at L1 and 0.980 at L2, against 0.712 and 0.662 for the
238 strongest model and 0.35–0.52 for the rest. The gap is therefore not an artifact of ambiguous or
239 unanswerable items: the information is in the released signals, and reading it out is what the panel
240 fails at. Level 3 behaves differently. There the expert reaches only 0.625 and does not lead the panel,
241 which we attribute to counterfactual prediction being poorly served by existing time-series tooling
242 rather than to the items being defective, and which means the headroom we report at L3 should be
243 read as smaller than at L1–L2.

244 5.3 Engineering decision-making at Level 4

245 **Level 4 exposes an industrial readiness gap.** On Level 4 the leader reaches 17.7% under the
246 free-form rubric and the other five score between 2.6% and 7.8% (recall that L4 uses an uncorrected
247 0/0.5/1 rubric and its magnitudes are not directly comparable to the chance-corrected L1–L3 scores;
248 the point of interest here is the rank-order within L4). Claude Sonnet 4.6, which leads L1–L3, ranks
249 near the bottom on L4 at 5.7%, alongside the weakest models in the panel. The model receives
250 the time series and a machine description but must produce the root-cause diagnosis and multi-step
251 recovery procedure from its own knowledge; the poor L4 performance indicates that even models with
252 strong L1–L3 signal comprehension either lack the machine-specific technical knowledge to diagnose
253 faults and prescribe remediation, or cannot ground that knowledge in the observed signals. Closing
254 this gap is the central challenge FactoryBench is designed to measure. We probe this representation-
255 versus-read-out question directly in Appendix L: linear probes on a frozen open-weight panel member
256 recover fault *presence* from the model’s activations well above chance (0.83) even where the model’s
257 own answer stays below chance — a *read-out* failure — whereas fault *type* is only weakly decodable
258 and no better than an untrained-network baseline, pointing to a representational gap rather than
259 merely a decoding one. Part of the L4 collapse is thus miscalibrated read-out, and part is missing
260 representation.

261 **GPT-5.1 leads on the most practically relevant level despite underperforming on the others.**
262 The L4 ordering reshuffles completely relative to L1–L3: GPT-5.1, only fourth of six on the structured
263 levels, leads L4 at 17.7%, more than twice the next-best. L4 requires naming the root cause and pro-
264 ducing a multi-step remediation procedure grounded in manufacturer documentation; we conjecture
265 that GPT-5.1’s reversal reflects disproportionate exposure to structured industrial documentation in
266 pretraining, letting it match a protocol to a fault description even when its signal comprehension
267 is weak. The pattern reveals a dissociation between *signal comprehension* and *protocol retrieval*:
268 optimizing for one does not deliver the other.

269 Within Level 4, GPT-5.1 shows asymmetric failure modes across troubleshooting and optimization.

270 Breaking GPT-5.1’s L4 performance down by question type (Figure 3b) exposes a structural
271 difference in *how* the model fails. On troubleshooting ($n=1,011$, mean 0.174) the score distribution is
272 sharply bimodal: 79.6% zero, 14.4% perfect, only 5.9% partial. The perfect-score mass concentrates
273 on *nominal* (no-fault) episodes (Figure 3b, left): GPT-5.1 emits “no anomaly” correctly on 33% of
274 healthy episodes, accounting for 92% of all perfects. The remaining 11 perfects fall on collision,
275 payload, and screw-thread faults whose signal signature is strong and unmistakable: a TCP force
276 spike, a speed-scaling collapse, or a large torque drift. Subtler anomalies (task-phase sequencing
277 errors like an unintended loosening step, or flag-level state changes) produce no catastrophic event;
278 GPT-5.1 then over-interprets the signal, attributing the change to a software bug or controller race
279 rather than the expected phase transition, and scores zero or 0.5. Large- n fault categories (loosening
280 phase, missing screw, extra component) sit at $\leq 11\%$ partial and $\leq 1\%$ perfect: even when GPT-5.1
281 senses that something is off, it cannot recover the canonical root cause from the signal.

282 On optimization ($n=102$, mean 0.201) GPT-5.1 *never* achieves a perfect score (41 partial, 61 zero;
283 Figure 3b, right). Ground-truth optimization answers are single-parameter corrections (e.g. “payload
284 mass is set to 0.0 kg but the actual payload weighs 1.5 kg; update the installation settings”). GPT-
285 5.1 instead emits lists of 8–12 generic motion-tuning suggestions (reduce acceleration, retune PID
286 gains, add waypoints, lower speed scaling, enable force-mode compliance) and mentions payload
287 configuration only as one undifferentiated item without specifying direction or magnitude, earning
288 0.5 at best (according to our grading policy). Together these patterns show that even though GPT-5.1
289 outperforms every other model by a wide margin on L4, it remains weak at decision-making in an
290 industrial context.

291 5.4 Modular and extensible by design

292 FactoryBench is built to grow. The Q&A generator decouples reasoning templates from the underlying
293 robot, task, and signal schema: every template is parameterized over generic primitives (signal names,
294 phase indices, anomaly labels, fault categories) rather than hard-coded to a machine, so adding a
295 new platform only requires populating the labeling ontology and per-robot signal mapping. All
296 existing templates then instantiate automatically, and the format scorers, knowledge graph, and
297 judge prompts carry over. Planned extensions include further robots and machine families (industrial
298 arms, mobile manipulators, CNC and additive-manufacturing platforms), tasks (assembly variants,
299 polishing, inspection), gripper and end-effector types, and templates targeting under-represented
300 reasoning patterns. Each addition slots into the existing schema without breaking prior versions;
301 the versioned release track pins every reported number to a fixed dataset state while the benchmark
302 continues to grow.

303 5.5 Limitations

304 FactoryBench has three substantive limitations. First, Level 3 *counterfactual* ground truth on physical
305 hardware is fundamentally approximate and resource-intensive to generate: two real runs cannot share
306 an identical pre-injection state, so our signature-kernel-MMD-selected CF pair is the closest empirical
307 proxy for the do-operation rather than the do-operation itself, and L3 scores therefore conflate model
308 reasoning with proxy quality (simulation closes the gap only at the cost of physical realism). Second,
309 as the first benchmark of its sort our Q&A pairs use a simplified fault model: industrial faults in
310 production are typically compound, gradual, and detected through aggregate KPIs, while ours are
311 atomic, fast, and drawn from a closed catalogue of 27 injected mechanisms, making FactoryBench a
312 measurement of in-distribution fault *recognition* rather than the fault *detection* problem operators
313 actually face. Third, Level 4 ground truth is drawn from a finite catalogue of manufacturer error codes
314 and expert-authored recovery procedures, so a model is effectively evaluated on *closed-book retrieval*
315 of protocols it may or may not have seen during pretraining rather than on open-ended engineering
316 reasoning; the low L4 scores we report therefore conflate “does the model know this specific vendor
317 manual?” with “can it reason from signals to a corrective action?”, and the two failure modes can
318 only be separated by controlling for the model’s prior exposure to the underlying documentation. As
319 a first step toward separating representational absence from read-out failure, Appendix L uses linear
320 probing to show that at least fault presence and fault type are linearly encoded in an open-weight
321 model’s activations even where its answers collapse to chance, isolating part of the behavioural gap
322 at the read-out rather than the representation.

6 Conclusion

We introduced FactoryBench, a four-level benchmark for machine understanding over industrial robotic telemetry, grounded in FactoryWave and structured around Pearl’s causal hierarchy. Zero-shot evaluation of six frontier LLMs shows that signal comprehension and protocol-grounded decision-making are distinct capabilities that do not co-develop: the panel ordering on Levels 1–3 reshuffles on Level 4, with GPT-5.1 taking the L4 lead despite being mediocre on the structured levels while the L1–L3 leader ranks near the bottom on L4. Structured-level scores remain well below 50% (chance-corrected) and no model approaches saturation on any level. To probe whether the gap can be closed by composition rather than scale, we build a tool-augmented LLM agent that delegates quantitative subtasks to specialised tools (a time-series foundation model, a sandboxed numerical interpreter, signal statistics, and vendor-manual retrieval) and reserve the LLM for the linguistic, classification, and decision-making templates where it has the structural advantage [30–32]; it lifts GPT-5.1 by +22.9, +13.8, +6.6, and +17.0 pp on L1–L4 respectively (Appendix I), but even the tool-augmented composition remains short of Claude Sonnet 4.6 on L1–L3, indicating that the remaining gap is not only a matter of tool availability. The gap is structural, not incidental: until models can read industrial signals, reason causally about them, and translate that reasoning into operator-grade decisions, they have no business on the factory floor. FactoryBench draws that line, and gives the community the instrument to measure every step taken toward crossing it.

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A Answer formats and scoring

FactoryBench emits questions in five answer formats. The four structured formats (single-select MCQ, multi-select MCQ, ranking, tensor) are scored deterministically by per-format rules; free-form answers, used only at Level 4, are scored by an LLM-as-judge voting protocol whose inter-judge reliability we analyse in Appendix A.1. Each format is designed to remain hard to answer in the absence of correct reasoning over the time series. Table 4 summarises the raw scoring rules; the rest of this section gives the implementation details together with the chance-correction step that places every format on a comparable scale before aggregation.

Table 4: Answer formats used in FactoryBench and their raw scoring rules. The chance-correction step described in this section is applied on top of these rules before per-level aggregation. Free-form answers (Level 4 only) are scored by an LLM-as-judge voting protocol with three judges (GPT-5.1, Claude Sonnet 4.6, DeepSeek V3.2) aggregated by the median of the three votes.

Format	Description	Raw scoring (before chance correction)
Single-select MCQ	Three or four mutually exclusive options (A–C or A–D); the model selects one.	Exact match: 1 if the parsed letter equals the gold letter, 0 otherwise. Per-item chance level $E = 1/k$ where $k \in \{3, 4\}$ is the option count for that question.
Multi-select MCQ	Four independent statements (A–D) about the outcome of an event or intervention; the model emits a T/F string of length four.	Positional fraction: $1/4$ per slot whose T/F matches the gold; raw score in $\{0, 0.25, 0.5, 0.75, 1\}$. Chance level $E = 1/2$.
Ranking	Four time-series segments or outcomes (A–D) ordered by a specified criterion. Answer is a permutation string (e.g. DABC).	Positional fraction: $1/4$ per slot whose letter matches the gold permutation; raw score in $\{0, 0.25, 0.5, 0.75, 1\}$. Chance level $E = 1/4$.
Tensor	One or more scalar values (e.g. expected sensor reading after an intervention). Answer is a list of floats.	Per-scalar three-level piecewise score: 1 if $ \hat{v} - v^* \leq m$, 0.5 if $m < \hat{v} - v^* \leq 2m$, else 0. Per-channel margin calibrated to $m_j = R_j/12$ so the chance level matches MCQ ($E = 1/4$).
Free-form	Open-ended natural-language response, used only for the two Level 4 question types: troubleshooting and optimization.	Score $\in \{0, 0.5, 1\}$ by LLM-as-judge against the reference answer: 1.0 if semantically equivalent to the reference, 0.5 if partially correct or imprecise but on-topic, 0.0 if incorrect, irrelevant, or refusing. The verbatim judge prompt is given in Appendix A.

Chance correction. Without correction, formats with different chance baselines are not directly comparable: a model scoring 0.5 on a multi-select item (chance $E = 1/2$) has demonstrated nothing, while a model scoring 0.5 on a single-select item (chance $E = 1/4$) has demonstrated above-chance performance. We therefore transform every raw item score s into a chance-corrected score

$$\tilde{s} = \frac{s - E}{1 - E},$$

where E is the format’s chance level. By linearity of expectation, uniform random guessing produces $\mathbb{E}[\tilde{s}] = 0$ for every format by construction; $\tilde{s} = 1$ corresponds to a perfect answer. The transform is signed: items scoring below chance receive $\tilde{s} < 0$, so a model that is worse than uniform random is visibly penalised in the aggregate mean rather than hidden by a floor. The per-format floor is $\tilde{s}_{\min} = -E/(1 - E)$, which ranges from -1 for multi-select ($E = 1/2$) to $-1/3$ for tensor / ranking / four-way MCQ ($E = 1/4$); aggregate means below zero should therefore be interpreted with the level’s format mix in mind. The non-LLM baseline used in Section 5.1 doubles as the empirical audit: after chance correction it hovers near 0 on every format (+1.9%, -3.0%, +1.7% on Levels 1–3 respectively), which we verify before treating any model score as evidence of reasoning.

Single-select MCQ. Raw score $s \in \{0, 1\}$. The chance level is read off the question payload as $E = 1/k$. The corrected score is $\tilde{s} = (ks - 1)/(k - 1)$, mapping a correct answer to $\tilde{s} = 1$ and a wrong answer to $\tilde{s} = -1/(k - 1)$ (i.e. $-1/2$ for $k = 3$, $-1/3$ for $k = 4$).

Multi-select MCQ. Raw score $s \in \{0, 0.25, 0.5, 0.75, 1\}$ with chance level $E = 1/2$ (each T/F slot matches the gold with probability $1/2$ under uniform random guessing). The corrected score is $\tilde{s} = 2s - 1$, mapping $\{0, 0.25, 0.5, 0.75, 1\}$ to $\{-1, -0.5, 0, 0.5, 1\}$.

Ranking. Raw score is the positional-match fraction over a permutation of length four. The expected number of fixed points of a uniformly random permutation is exactly 1 regardless of length, so the chance level is $E = 1/n = 1/4$. The corrected score is $\tilde{s} = (4s - 1)/3$, mapping $\{0, 0.25, 0.5, 0.75, 1\}$ to $\{-1/3, 0, 1/3, 2/3, 1\}$.

Tensor. Each scalar j in a vector answer is scored on a three-level scale: 1 if $|\hat{v}_j - v_j^*| \leq m_j$, 0.5 if within $2m_j$, 0 otherwise. The raw item score is the average over the n scalars, $s = \frac{1}{n} \sum_j s_j$. The discrete form mirrors MCQ/ranking and still credits near-misses inside $2m_j$. The per-channel margin is calibrated to $m_j = R_j/12$, where R_j is the channel’s empirical $[p_2, p_{98}]$ range across the unified episodes (precomputed once); this sets the piecewise scorer’s chance expectation to $E = 1/4$, matching single-select MCQ. The corrected item score follows the MCQ form, $\tilde{s} = (4s - 1)/3$. Scalars with a degenerate channel range ($R_j \approx 0$) are dropped from the average.

Free-form. Free-form answers do not admit a meaningful chance baseline: the answer space is unbounded natural language, so there is no uniform random distribution to subtract. We therefore leave free-form scores uncorrected. The lack of correction does not introduce cross-format bias because free-form is used *only* at Level 4, and Level 4 contains no other answer format, so free-form scores are never aggregated alongside structured-format scores at the same level. As a consequence, Level 4 scores and Levels 1–3 scores are reported on different underlying scales (an uncorrected 0/0.5/1 rubric versus a chance-corrected structured metric) and are *not directly comparable in magnitude*; when we contrast them in Section 5.1 we refer to rank-order and gap-to-baseline effects rather than absolute-difference claims.

Every free-form answer is scored by the LLM judge against the gold reference answer (assembled from the FactoryBench knowledge graph) using the prompt below, reproduced *verbatim* from the scoring code and applied identically to both Level 4 question types (troubleshooting and optimization). The placeholders {question}, {reference}, and {prediction} are filled per item:

```

552 You are an expert evaluator for a robotics sensor-data Q&A benchmark.
553 Score a model’s free-form answer against a reference answer.
554
555 Scoring:
556   1.0 (Correct):  semantically equivalent to the reference: it gives the correct
557                   root cause / misconfigured parameter AND a plausible corrective
558                   step (exact wording of the fix is NOT required).
559   0.5 (Neutral):  on-topic AND it identifies the correct root cause / misconfigured
560                   parameter, even if imprecise or without the corrective step.
561   0.0 (Wrong):    incorrect, irrelevant, names the WRONG root cause, is a generic
562                   list that MISSES the specific parameter, claims a fault when the
563                   reference says the machine is normal (or vice versa), or refuses
564                   to answer.
565
566 Being merely "on-topic" is NOT enough for 0.5: the answer must identify the actual
567 root cause / parameter from the reference. A generic list of tuning suggestions
568 that does not single out the reference’s specific cause/parameter is 0.0.
569
570 Respond with ONLY a JSON object on a single line, no markdown:
571 {"score": <0.0|0.5|1.0>, "reason": "<one short sentence>"}
572
573 Question:
574 {question}
575
576 Reference answer:
577 {reference}
578
579 Model answer:
580 {prediction}

```

581 A held-out audit confirms the panel applies this rubric consistently: an independent grader (outside
 582 the three-judge panel) given this exact prompt matches the judge-median on 95% of a representative
 583 100-item sample (quadratic-weighted $\kappa = 0.79$).

584 **Aggregation.** Per-question chance-corrected scores $\tilde{s}$ are averaged uniformly within a template and
 585 level.

586 A.1 Inter-judge reliability of the Level-4 panel

587 Because free-form Level-4 answers are scored by three judges (GPT-5.1, Claude Sonnet 4.6, DeepSeek
 588 V3.2) aggregated by median (Table 4), the L4 scores inherit whatever (dis)agreement those judges
 589 carry. We therefore quantify their reliability. We pool every Level-4 answer for which all three judges
 590 returned a valid verdict across the six-model evaluatee panel ($n=6,617$ answers, three verdicts each
 591 in $\{0, 0.5, 1\}$); inter-rater agreement is a property of the judges, so pooling across evaluatees is the
 592 appropriate unit of analysis.

593 **The panel is reliable.** Agreement is substantial-to-good: Fleiss’ $\kappa = 0.60$ and interval Krippen-
 594 dorff’s $\alpha = 0.78$ across the three raters, with 87.5% of items unanimous, 11.9% carrying a 2-of-3
 595 majority, and only 0.6% three-way splits (Figure 4b). Pairwise, exact agreement is 89.7–93.4%
 596 (mean 91.5%) and the quadratic-weighted κ is 0.74–0.81 (Figure 4a), with GPT-5.1 and DeepSeek
 597 agreeing most and Claude and DeepSeek least. The pairwise confusion matrices are strongly diagonal
 598 and disagreements are almost never polar ($0 \leftrightarrow 1$): they are overwhelmingly between adjacent grades
 599 ($0 \leftrightarrow 0.5$ or $0.5 \leftrightarrow 1$), consistent with the ordinal rubric.

600 **A small leniency spread justifies median aggregation.** The judges differ only mildly in strictness
 601 — mean verdict 0.101 (Claude), 0.082 (GPT-5.1), 0.066 (DeepSeek), so Claude is the most lenient
 602 and DeepSeek the strictest. This directional bias is visible in the off-diagonals of the confusion
 603 matrices (e.g. Claude assigns 0.5 where DeepSeek assigns 0 on 8% of items), and it is exactly the
 604 regime in which the *median* is preferable to the mean: it is robust to a single lenient or strict judge and
 605 returns the majority verdict whenever one exists, which it does for 99.4% of items. High agreement
 606 together with the robustness of median aggregation indicates that the reported Level-4 scores are not
 607 an artefact of any single judge.

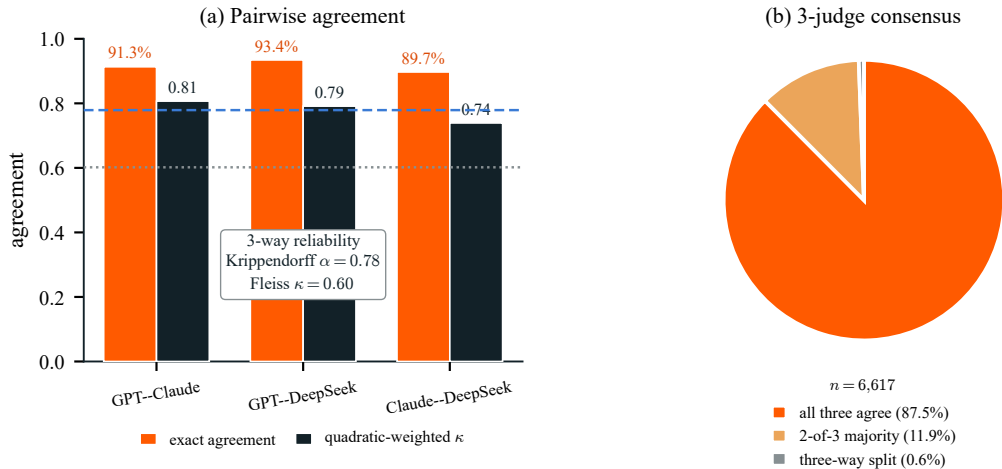


Figure 4: **The three-judge Level-4 panel is reliable.** (a) Pairwise exact agreement (91.3%, 93.4%, 89.7%; mean 91.5%) and quadratic-weighted κ for each judge pair, with the 3-way Fleiss $\kappa=0.60$ and Krippendorff $\alpha=0.78$ as reference. (b) Consensus across the three judges: all three agree on 87.5% of items, 11.9% are 2-of-3 majorities, and only 0.6% are three-way splits. Pooled over 6,617 Level-4 answers with three valid verdicts.

B Released Q&A distribution

Figure 5 summarises how the released 70,915 Q&A items are distributed across levels, answer formats, and time-series context lengths. Three patterns are worth flagging.

Level sizes are uneven by design. L2 (33,425 items) is by far the largest level and L3 (2,939 items) is the smallest. The asymmetry follows directly from the underlying episode pool. Most FactoryWave episodes are anomalous (we deliberately oversample fault scenarios so that diagnostic templates have enough material), and L2 templates can be instantiated against any anomalous episode, which inflates L2’s count. Counterfactual templates (L3) by contrast require paired (baseline, fault) runs collected with the signature-kernel-MMD protocol of Section 4.2, which is far more expensive to produce per episode; the released L3 corpus is therefore the slice of items for which a high-quality CF pair exists, and is intrinsically small.

Answer-format mix tracks the reasoning level. L1 leans on numerical (69 %) and single-select MCQ (28 %) formats; L2 spreads across single-select MCQ (48 %), numerical (41 %), and multi-select (7 %), with small ranking and tensor slices (2 % each); L3 is the most format-diverse level, splitting across multi-select (42 %), numerical (21 %), ranking (20 %), and tensor (17 %), reflecting the counterfactual ranking and trajectory-prediction templates; L4 is entirely free-form because every L4 template asks for natural-language remediation or optimisation answers. Free-form appears only at L4: no L1–L3 template asks for one.

Sub-series lengths are tightly bounded. The displayed time-series window is between 32 and 77 timesteps (>64 is possible for ranking since multiple time series) for almost every released item (median ≈ 50 across all four levels), keeping the prompt within the context budget of every evaluated model.

B.1 Question-template inventory

Table 5 lists the number of question templates currently producing released Q&A items, broken down by answer format. Templates that are defined in the generator but do not yet emit any questions in the released set are excluded.

Figure 6 instantiates Pearl’s three causal levels (Association, Intervention, Counterfactual) on robotic time-series data and adds the Level 4 decision-making layer that FactoryBench introduces on top. Each panel pairs the formal causal primitive with a representative FactoryBench template.

Table 5: Number of active question templates per level, by answer format. Counts reflect templates that produce released Q&A items in the current dataset.

Level	MC single-select	MC multi-select	Ranking	Tensor / Numerical	Free-form	Total
1 (State)	1	1	0	2	0	4
2 (Intervention)	3	2	2	3	0	10
3 (Counterfactual)	0	2	1	2	0	5
4 (Decision)	0	0	0	0	2	2
Total	4	5	3	7	2	21

Expert-authored severity ranking. Several Level-2 ranking templates (L2.1, L2.9) require ordering anomalies or signal segments by severity. Severity is not directly readable from the labels: it depends on physical impact, controller reaction, recovery cost, and downstream production consequences. The reference ordering used for these templates was authored and cross-checked by a panel of PhD-level robotics experts, who ranked each anomaly category along these axes and reviewed the resulting ranks for cross-anomaly consistency. The fixed expert ranking is shipped with the benchmark and used as the gold permutation by the deterministic positional-match scorer (Appendix A).

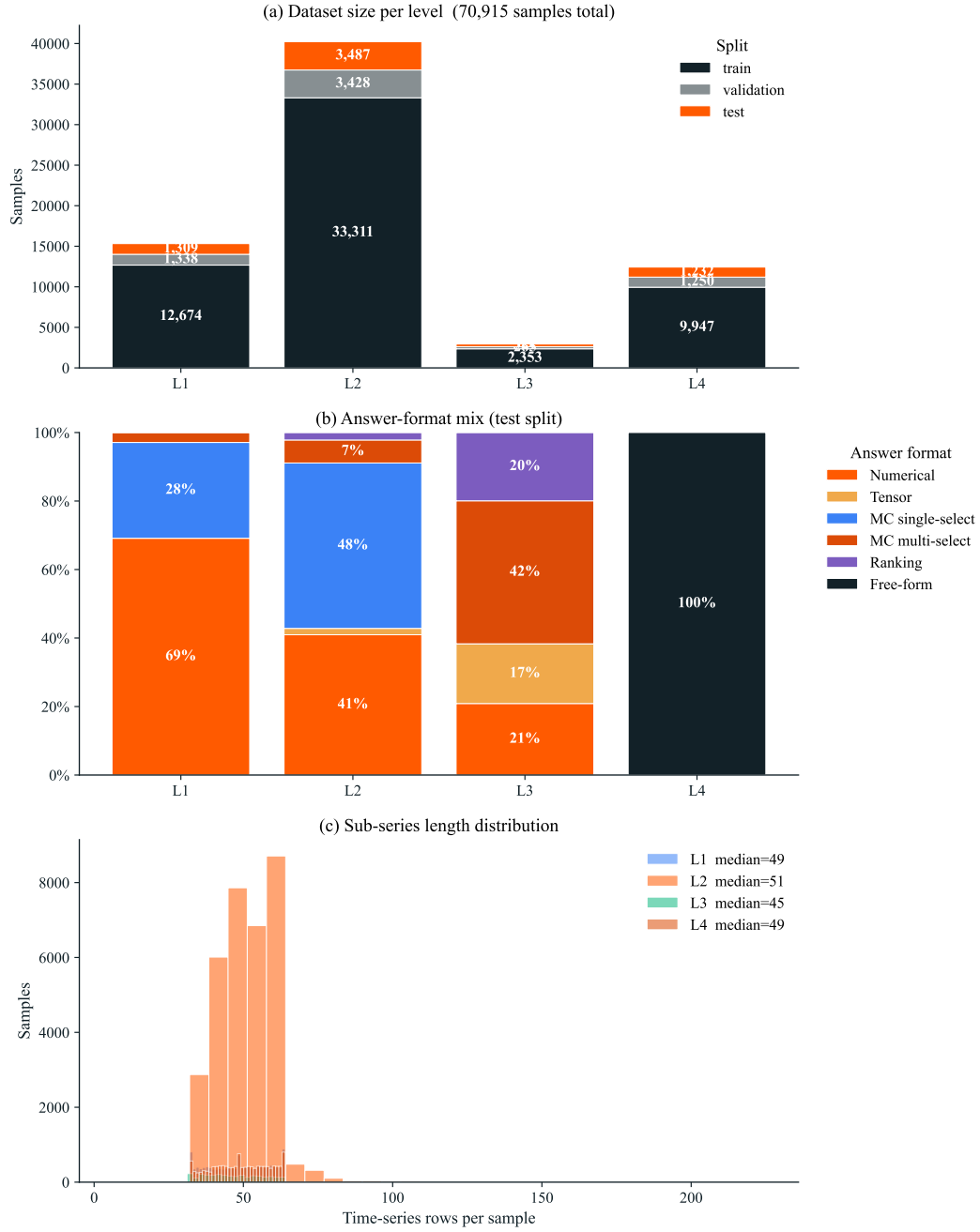


Figure 5: Distribution of the 70,915 released Q&A items. **(a)** Level sizes with train/validation/test breakdown. **(b)** Answer-format mix on the test split, per level. **(c)** Sub-series length (number of time-series rows) per item.

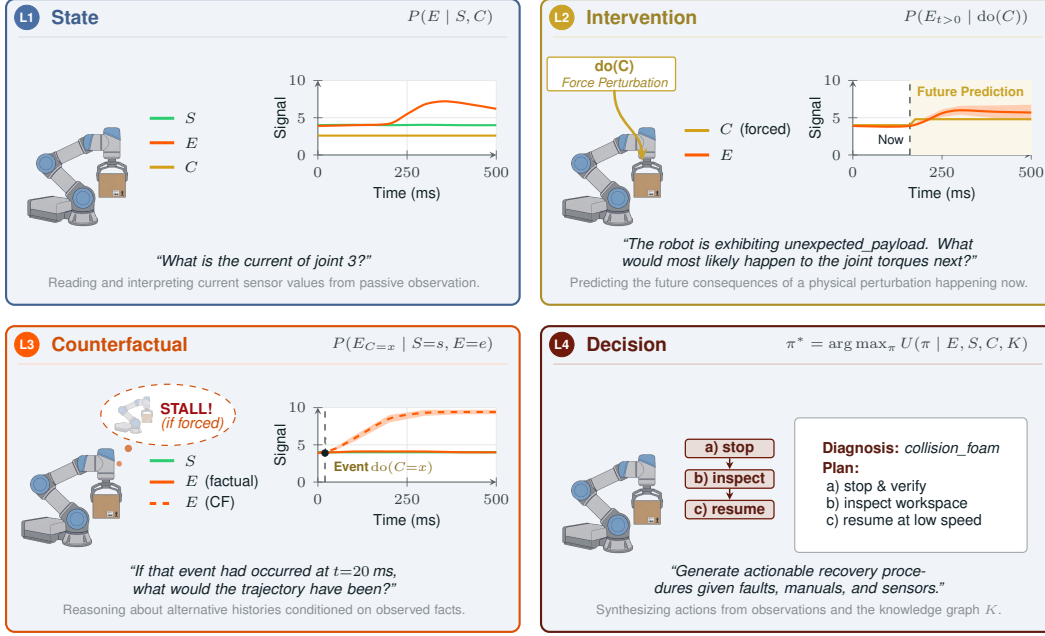


Figure 6: The four levels of machine understanding defined by FactoryBench.

B.2 Full template list

Table 6 lists every active template (the templates that emit released Q&A items in the current dataset). Placeholders in braces ($\{\text{task}\}$, $\{\text{anomaly}\}$, $\{\text{phase}\}$, $\{\text{signal}\}$, $\{n\}$, $\{\text{event}\}$, $\{\text{window_length}\}$, $\{\text{joint_signal}\}$) are filled at generation time from the episode metadata or the relevance specification. The trailing answer-format reminders (“*Answer only with . . .*”) are truncated for readability, matching the convention used in Table 19.

Table 6: Full list of FactoryBench question templates by causal level. *Single-choice MCQ* requires one letter; *multi-select MCQ* requires a binary string over a fixed option set; *ranking* requires a permutation over k items; *tensor (scalar)* is a single real number accepted within a tolerance band; *tensor (6-vector)* is a JSON array of six real numbers, one per joint axis; *free-form* responses are evaluated by an LLM-as-judge against a reference answer.

ID	Answer format	Template (boilerplate truncated)
Level 1: State		
L1.1	Tensor (scalar)	The robot is performing a $\{\text{task}\}$ task. We want to isolate the $\{\text{phase}\}$ in the robot’s time series. Assuming a fixed window length of $\{\text{window_length}\}$ timesteps, at which timestamp should the window begin?
L1.3	Multi-select MCQ	What changed between the two instances of robotic time series data?
L1.6	Single-choice MCQ	What robot does this sensor data originate from?
L1.7	Tensor (scalar)	Given the sensor stream below, what is the expected value of $\{\text{signal}\}$ at $T+\{n\}$ ms?
Level 2: Intervention		
L2.1	Ranking	The sensor stream below is from a robot exhibiting $\{\text{anomaly}\}$. Rank the signal segments listed in the ‘options’ field in the order you would expect them to appear as the anomaly manifests.

continued on next page

Table 6 continued from previous page

ID	Answer format	Template (boilerplate truncated)
L2.2	Multi-select MCQ	The sensor stream below is from a robot exhibiting {anomaly}. What would most likely happen next?
L2.3	Multi-select MCQ	The sensor stream below is from a robot exhibiting {anomaly}. Select all statements that apply in the options provided.
L2.4	Tensor (scalar)	The sensor stream below is from a robot exhibiting {anomaly}. What is the expected value of {signal} at T+{n}ms?
L2.5	Tensor (6-vector)	The sensor stream below is from a robot exhibiting {anomaly}. What are the expected values of {joint_signal} at T+{n}ms?
L2.6	Tensor (scalar)	Knowing that the robot suffers from {anomaly} in the given context time series, we want to isolate the {phase} phase. Assuming a fixed window length of {window_length} timesteps, at which timestamp should the window begin?
L2.7	Single-choice MCQ	Given the sensor data from a robot performing the task, determine what anomaly is present?
L2.8	Single-choice MCQ	You are provided with two sensor streams originating from robots accomplishing tasks. What differences between the two given instances of robotic time series data (if any) do you notice?
L2.9	Ranking	Rank the following robot time series segments from most to least severe anomaly.
L2.10	Single-choice MCQ	Knowing that the robot suffers from {anomaly} in the given context time series, what robot does this sensor data originate from?
Level 3: Counterfactual		
L3.1	Ranking	Given the baseline sensor stream below and the counterfactual scenario where {event}, rank the signal segments listed in the ‘options’ field in the order you would expect them to appear as the event manifests.
L3.2	Multi-select MCQ	Given the counterfactual scenario where {event}, what would most likely happen next?
L3.3	Multi-select MCQ	Given the counterfactual scenario where {event}, select all statements that would apply.
L3.4	Tensor (scalar)	In the counterfactual scenario where {event}, what would be the expected value of {signal} at T+{n}ms?
L3.5	Tensor (6-vector)	In the counterfactual scenario where {event}, what would be the expected values of {joint_signal} at T+{n}ms?
Level 4: Decision		
L4.1	Free-form	Given the sensor stream below, does the machine show signs of anomalous behavior? If yes, identify the most likely root cause and describe the steps you would take to fix it.
L4.2	Free-form	An engineer wants to increase the effectiveness and accuracy of this machine. Based on the sensor stream below, what operational changes or parameter adjustments could help achieve this? Do not justify your answer, simply indicate steps to take (or specify if nothing can be done).

651 **Prompt structure and knowledge-graph augmentation.** Models under evaluation receive a single
652 user message; no system role is set on the evaluated model. The message is assembled at prompt-build
653 time from five ordered blocks:

```
654 <machine_sentence>
655 <task_phase_reference>
656 <acronym_mapping>
657 <time_series>
```

658 Question: <question_text>
 659 Here are the options:
 660 <options>

661 The leading <machine_sentence> is filled at prompt-build time from the FactoryBench knowledge
 662 graph, specifically the machine and gripper tables, keyed on the dataset that the question is grounded
 663 in. This guarantees that identical question templates are grounded in the correct hardware description
 664 regardless of which underlying dataset the episode comes from. For a UR3e-grounded question on
 665 FactoryWave, the resulting line expands to:

666 The following sensor data comes from Universal Robots UR3e, a collaborative robot (cobot)
 667 from the E-series with 3 kg payload, 6 degrees of freedom, controlled via UR PolyScope
 668 (teach pendant), UrScript, typically used for light assembly tasks, automated workbench
 669 scenarios. It is equipped with a 2FG14 Finger Gripper OnRobot.

670 The augmentation is conditional: identification templates such as L1.6 (“What robot does this sensor
 671 data originate from?”) and L2.10 suppress the machine sentence so the model cannot read the answer
 672 off the prompt header, and templates that probe gripping behavior without revealing tooling can
 673 selectively suppress only the gripper clause. The acronym-mapping block decodes the time-series
 674 header into long-form channel names and is included whenever the question carries one.

675 The <task_phase_reference> block defines the task’s phases, and is attached to every prompt
 676 whose question or options refer to a phase, or whose context exposes the task_phase channel.
 677 Without it the phase vocabulary is ambiguous in two distinct ways: the task_phase channel appears
 678 in the series as a bare integer with no legend, and the question templates refer to phases in prose (“the
 679 grasp of the object phase”) that does not match the canonical phase names in the knowledge graph.
 680 The block resolves both by listing the phases of the episode’s task in their fixed execution order,
 681 giving for each the canonical name, the wording the templates use, and a one-line description, with
 682 the integer label included only when the context actually exposes task_phase. For a pick-and-place
 683 episode it expands to:

684 Task: Pick and Place. Robot picks an object from a source location and places it at a target
 685 location (bin). The task runs through the following phases in this fixed order, labelled by
 686 the task_phase signal: 0 = Above Pick (referred to in questions as “approach to the
 687 object”): EEf moves to a waypoint directly above the object. . . 3 = Close (referred to in
 688 questions as “grasp of the object”): Gripper closes around the object. . .

689 The block is deliberately timing-free: it states what a phase *is* and where it sits in the task’s fixed
 690 order, never which timesteps of the episode at hand belong to it. That distinction matters because
 691 several templates (L2.6 in particular) ask the model to locate a phase boundary inside the displayed
 692 window, and naming the timesteps would hand over the answer. Phase definitions are shared context;
 693 phase segmentation is the thing under test.

694 Level-4 free-form responses are scored by an LLM-as-judge under a strict three-point rubric
 695 ({0.0, 0.5, 1.0}) that is template-specific: troubleshooting items (template L4.1) and optimiza-
 696 tion items (template L4.2) use different prompts, since the relevant axes of correctness differ (root
 697 cause + remediation steps vs. problematic parameter + adjustment direction).

698 **Troubleshooting prompt (L4.1).**

699 You are scoring a model’s troubleshooting answer for a robotics
 700 sensor-data benchmark.

701

702 Rubric (apply strictly, no other values allowed):

703 1.0 - the answer correctly identifies the underlying root cause AND
 704 the proposed remediation steps are mostly right. The steps do
 705 NOT need to match the reference exactly: minor wording
 706 differences, additional sensible checks, or omitted minor
 707 steps are all fine, as long as the overall remediation is on
 708 the right track and would plausibly resolve the issue.

709 0.5 - the answer correctly identifies the underlying root cause but
 710 the remediation is wrong, missing, or off-topic (e.g. proposes

711 an unrelated fix, contradicts the reference, or refuses to
 712 give steps).
 713 0.0 - the root cause is wrong or absent (model refused, identified
 714 the wrong fault, or stated "no anomaly" when one was present).

715
 716 The known root cause is provided to you separately. An answer counts
 717 as identifying the root cause when it names the same physical
 718 phenomenon - equivalent paraphrases are fine, but a different fault
 719 category (e.g. saying "payload too heavy" when the root cause is
 720 "TCP misconfiguration") is wrong.

721
 722 Respond with ONLY a JSON object on a single line, no markdown:
 723 {"score": <0.0|0.5|1.0>, "reason": "<one short sentence>"}

724
 725 Question: {question}
 726 Reference: {reference}
 727 Known root cause:{root_cause}
 728 Model answer: {prediction}

729 Optimization prompt (L4.2).

730 You are scoring a model’s optimization answer for a robotics sensor-
 731 data benchmark.

732
 733 Rubric (apply strictly, no other values allowed):

734 1.0 - the answer identifies the problematic parameter AND proposes
 735 adjusting it in the correct direction (increase vs. decrease,
 736 matching the reference). Exact magnitudes do not matter; only
 737 direction matters.
 738 0.5 - the answer identifies the problematic parameter but proposes
 739 the wrong direction, no direction, or contradictory directions.
 740 0.0 - the problematic parameter is not identified, or the model
 741 recommends adjusting an unrelated parameter, or refuses to
 742 answer.

743
 744 The reference answer states the correct parameter and its target
 745 value, from which the correct adjustment direction can be inferred
 746 relative to the configured value.

747
 748 Respond with ONLY a JSON object on a single line, no markdown:
 749 {"score": <0.0|0.5|1.0>, "reason": "<one short sentence>"}

750
 751 Question: {question}
 752 Reference: {reference}
 753 Model answer:{prediction}

754 The judge ensemble comprises three frontier LLMs from different vendors: **GPT-5.1**, **Claude Sonnet**
 755 **4.6**, and **DeepSeek V3.2**. Each judge independently scores every Level-4 free-form response with
 756 the appropriate template-specific prompt above, and the per-item score reported in our results is the
 757 median of the three valid votes (snapped to the {0, 0.5, 1} grid).

758 C Episode eligibility filtering

759 Before any sub-window is drawn (Appendix D), each template restricts the pool of source episodes to
 760 those that can support a well-defined ground-truth answer. The filter is template-aware and combines
 761 a level-specific condition predicate with a length/coverage check.

762 **Level-specific condition predicate.** The level a template lives at fixes which episode condition the
 763 template can pull from: Level 1 templates probe nominal state and therefore consume only *baseline*

(normal) episodes, Level 2 templates probe the immediate effect of an intervention and therefore consume only *anomalous* (fault) episodes, and Level 3 templates probe counterfactual reasoning and therefore consume only *counterfactual* pairs (one baseline plus one fault run, paired by the protocol described in Appendix G). Level 4 templates draw from baseline and fault episodes depending on the troubleshooting/optimization variant. Episodes whose condition does not match the template are removed from the pool before sampling.

Ground-truth feasibility. Eligible episodes must additionally carry the labels and trajectory length the template needs to derive its answer. Concretely, we require that (i) the episode contains every metadata field the template instantiates (root-cause label, anomaly type, fault-injection timestep, target phase, etc.), (ii) the episode is long enough that a relevance-compliant sub-window of the requested length (Appendix E) fits, and (iii) for templates whose answer is computed from a future segment of the trajectory (numerical and tensor forecasts at L1.7, L2.4, L2.5), the trajectory extends at least the requested forecast horizon beyond the displayed window. Episodes that fail any of these checks are dropped at filter time rather than at sampling time, so each template’s sample distribution is over its true eligibility pool.

Effect on the released set. The filter is applied per (template, dataset, condition) cell before the train/validation/test split, so the public 70k Q&A items released on Hugging Face are exactly the items for which a verifiable ground-truth answer exists.

D Fault-relevance window sampling

For Q&A pairs grounded in a fault episode, the time-series context shown to the model is a sub-window of the full episode rather than the entire trajectory. A naive uniformly sampled sub-window can miss the segment where the fault is actually visible: for a transient collision lasting tens of milliseconds, a uniformly sampled one-second window has a non-negligible probability of falling entirely outside the disturbed segment, producing an item whose ground-truth label is correct but whose evidence is invisible to any model. To prevent this, each fault type is paired with a *relevance specification* that constrains how the sub-window is drawn so that the fault signature is, by construction, present in the displayed context.

Four temporal footprints. Each fault is assigned, by domain annotation, to one of four classes that describe how its signature is distributed across the episode:

- *Global*: the signature is present throughout the entire episode, for instance a payload misconfiguration that biases every torque reading. Sub-windows are sampled uniformly under the requested length bounds.
- *Event-anchored*: the signature is a transient localized at a specific timestep recorded in the episode (collisions, gripper misactivation). The sub-window is required to contain that timestep; its length and start are otherwise random.
- *Phase-gated*: the signature is only diagnostic during specific task phases, for instance a peg-misalignment fault that is only visible during insertion. The sub-window is required to overlap one of the target phases by at least a minimum number of rows, where the relevant phases depend on the task.
- *Cumulative*: the signature has a monotonically growing footprint that becomes visible only after a certain phase begins, for instance progressive torque drift. The sub-window is required to be at least a minimum length and, where applicable, to start no earlier than the onset phase.

Sampling and validation. Given a fault episode and the requested length bounds, the sampler draws a window that satisfies the corresponding constraint and records the constraint that was applied alongside the question’s provenance, so downstream analyses can condition on it. A post-hoc check verifies that the returned window meets the locality’s requirement; if it cannot be met (e.g., a phase-gated fault whose target phase is missing from the recording), the item is either discarded or, for templates where the anomaly signature is helpful but not required for correctness, the sub-window falls back to uniform sampling and the item is flagged accordingly.

Effect on the dataset. Relevance-aware sampling is applied to all fault-grounded items in Levels 1, 2, and 4. Level 3 counterfactual pairs use a separate fixed-injection-timestep protocol described in Section 4.2 and do not rely on this mechanism. In pilot generation runs without relevance constraints, a non-trivial fraction of fault items had sub-windows that did not contain the fault signature, inflating the apparent difficulty of Levels 1 and 2 in a way that could not be distinguished from genuine model failure. The mechanism can be disabled to recover uniform sampling for ablation studies that require a no-relevance baseline.

E Time-series context: features and window length

Two design choices govern what each Q&A item actually shows the model: *which* channels of the multivariate trajectory are included, and *how long* the displayed window is. Both choices are deliberately constrained.

Resampling to a uniform 10 Hz. The source datasets record at heterogeneous frequencies (83–125 Hz for FactoryWave, 100 Hz for AURSAD [1], 100 or 500 Hz for voraus-AD [2]). Before any Q&A item is built, every episode is anti-alias filtered and its *context window* is downsampled to a common nominal 10 Hz timeline. This applies to the rows the model sees in the prompt; the `timestamp_ms` column is preserved from the nearest source-rate sample and therefore is not exactly on a 100 ms grid (consecutive row offsets typically fall in [95, 105] ms depending on the native rate). Post-event windows used to compute forecast ground truth (Levels 2 and 3, templates 4 and 5) additionally retain the native sampling rate to preserve fast transients around fault onset; the answer horizon can therefore be sub-100 ms even though the visible context rows are ~ 100 ms apart. This context-downsampling serves three purposes: it normalizes the time axis across robots so that templates can specify window lengths in absolute seconds without conditioning on the source platform, it keeps the displayed context within the prompt budget of every evaluated model, and it removes high-frequency content that would otherwise dominate the token sequence with information not needed for the reasoning levels we evaluate. Ten Hertz is admittedly low by industrial-sensor standards: at higher sampling rates one can resolve fast transients (millisecond-scale collisions, valve switches) that are smoothed away here. We accept this trade-off because the fault catalogue evaluated in this paper is deliberately simplified to atomic, well-isolated mechanisms, and we observe empirically that 10 Hz is sufficient to expose every fault signature in our catalogue at a level that PhD-level annotators can verify on the displayed context. Future versions of the benchmark targeting compound or sub-second faults would need to revisit this choice.

Per-template feature pre-selection. The raw signal schema is large: FactoryWave exposes more than 120 channels per timestep (joint positions, velocities, accelerations, currents, target torques, contact forces, TCP pose, gripper state, and so on). Including every channel verbatim would produce contexts that are both prohibitively long and dominated by signals irrelevant to the question being asked. We therefore equip every question template with a hand-curated subset of *important features*: 30 channels per template for Levels 1–3, and roughly 20 for Level 4 (whose templates focus on the diagnostic and remediation signals that matter for troubleshooting). The selections are authored by PhD-level robotics experts on the team and target the channels most informative for the specific reasoning skill the template probes; for instance, an anomaly-detection template emphasizes torques and contact forces, while a phase-isolation template foregrounds joint positions and velocities. The full per-template feature lists are released alongside the benchmark so that every Q&A item can be reproduced bit-for-bit from the underlying episode.

Window length. Across all four levels, each Q&A item shows a sub-window of between 32 and 64 timesteps drawn from the 10 Hz-resampled trajectory, corresponding to roughly 3.2–6.4 s of robot motion. The lower bound is set so that even the shortest displayed window contains enough samples to expose the dynamics the templates probe (transient onsets, phase boundaries, force spikes); the upper bound keeps the prompt within the context limits of all evaluated models, including the smaller open-weight ones. Within these bounds, the exact length and start position are randomized at generation time, with the start position constrained by the relevance specification of Appendix D when the item is fault-grounded. These bounds apply per displayed window rather than per item, so templates that show more than one window carry a proportionally longer total context: comparison

866 templates present two sub-windows side by side, and ranking templates add four candidate segments
867 on top of the context window.

868 **Why these choices matter for interpretability.** Fixing the resampling rate, the feature subset, and
869 the window-length range across templates means that performance differences between models on a
870 given template cannot be attributed to one model receiving a longer, faster, or richer context than
871 another, only to its ability to interpret the same content. The randomization within bounds prevents
872 models from exploiting positional or length cues to recognize templates by their surface form: two
873 items instantiated from the same template are typically presented with different lengths, different
874 start positions, and – once relevance constraints and per-template feature subsets are honoured – with
875 the fault evidence falling at different relative positions in the displayed window.

876 F Validation of solvability

877 This appendix collects the diagnostic checks we ran to convince ourselves that FactoryBench items
878 are answerable from the released time-series context, rather than from question wording alone, and
879 that the templates flagged as “too easy” by the noise-substitution check are nevertheless grounded in
880 genuine signal structure.

881 F.1 Noise-substitution check

882 To probe whether the model scores in Figure 3 reflect actual reading of the time-series context
883 or merely priors over question templates and option positions, we constructed a noise-substituted
884 counterpart of the dataset and re-ran every model against it. This check was an early diagnostic study
885 run on a previous, smaller model panel (Claude Haiku 4.5, DeepSeek V3.1, gpt-5.1, Mistral Large
886 3) and a 400-item Q&A subset (100 items per level); it predates the main evaluation reported in the
887 body of the paper, so the absolute scores below are not directly comparable to those in Figure 3.
888 In particular, this early subset was generated with substantially longer time-series contexts than
889 the released benchmark, which depresses scores across the board for every model in the panel; the
890 relative differences between Orig and Sub remain interpretable, but absolute levels sit well below the
891 comparable cells in the main results.

892 **Construction.** For each of the 400 generated Q&A pairs (100 per level), we duplicate the item
893 and rewrite only the numerical time-series content. For each numeric *float* feature we preserve the
894 original first value and substitute every subsequent value with the cumulative sum of independent
895 Gaussian increments:

$$v'_0 = v_0, \quad v'_{t+1} = v'_t + \mathcal{N}(0, \sigma^2), \quad \sigma = 0.5,$$

896 where σ is shared across all features. The same substitution is applied to time-series chunks embedded
897 in option strings (Level 1 severity-ranking, Level 2/3 chunk-ranking).

898 **Expected behavior.** A model that scores by genuinely reading the time series should drop sharply
899 on the noise-substituted variant, because the substitution destroys the signal-trajectory information
900 that the gold answer hinges on. A model that scores by template-level priors (e.g. guessing the
901 most plausible MC option given the question wording) or by option-position bias should be roughly
902 invariant, since the question text and options are unchanged.

903 **Result.** Table 7 reports the side-by-side mean signed chance-corrected scores. Three patterns
904 emerge.

- 905 1. On Level 1 every model is essentially flat or improves slightly under substitution ($\Delta \in$
906 $[-0.8, +4.0]$). This is consistent with L1 templates being substantially answerable from
907 the question wording, the option set, and the still-present discrete annotations of task phase
908 and fault label. It also flags L1 as the level where overlap between LLM scores and a
909 random/regression baseline is most expected.
- 910 2. On Level 3 every model drops sharply, by 8 to 18 signed CC points; on Level 2 the two
911 strongest signal-readers of the panel (gpt-5.1 and DeepSeek V3.1) also drop substantially
912 (-12.0 and -3.4 respectively) while the weaker readers stay flat. The L3 drops are the most

informative: L3 templates ask counterfactual “what would have happened” questions where the answer hinges on the actual signal trajectory rather than on a discrete annotation, and every panel member’s L3 score collapses from clearly-above-chance to at-or-below-chance under substitution.

3. On Level 4, gpt-5.1’s 7.6% original score collapses to 0.0% under noise substitution. Inspection of the per-item rubric notes shows that gpt-5.1’s L4 mass on the original data came almost entirely from correctly emitting “no anomaly is present” on truly nominal episodes; once the time series is replaced by Gaussian noise, the same nominal episodes look anomalous to the model and it now hallucinates faults on them, losing its only mode of scoring. The other three models score ≤ 0.005 in either condition and have no comparable mode to lose.

Table 7: Noise-substitution check: signed chance-corrected mean scores (%) on the original FactoryBench Q&A pairs (Orig) vs. the noise-substituted variant (Sub) constructed by replacing every numerical time-series value with cumulative Gaussian increments seeded at the original first value ($\sigma = 0.5$). L4 uses the uncorrected free-form rubric. Questions, options, and gold answers are identical between the two runs. $\Delta = \text{Sub} - \text{Orig}$; large negative values are evidence that the model was using the data from the time series rather than the question alone.

Method	L1			L2			L3			L4		
	Orig	Sub	Δ	Orig	Sub	Δ	Orig	Sub	Δ	Orig	Sub	Δ
Claude Haiku 4.5	-12.7	-8.7	+4.0	+2.1	+4.5	+2.4	+8.6	-5.6	-14.2	0.5	0.0	-0.5
DeepSeek V3.1	-15.7	-16.5	-0.8	-1.7	-5.1	-3.4	+5.3	-2.8	-8.1	3.5	0.0	-3.5
gpt-5.1	+2.8	+3.5	+0.7	+7.2	-4.8	-12.0	+12.6	-2.6	-15.2	7.6	0.0	-7.6
Mistral-Large-3	-1.2	+2.5	+3.7	-0.7	-1.6	-0.9	+17.9	-0.2	-18.1	0.0	0.0	0.0

F.2 Human expert baseline

The strongest evidence that an item is solvable from the released telemetry is a human solving it. We therefore had a robotics expert answer a 60-item sample spanning Levels 1–3 under open-tool, no-LLM conditions, and scored the answers with the benchmark’s own grader.

Protocol. *Open tools.* The annotator was free to use any analysis instrumentation, and built a small Python toolkit for the task: series parsing, windowed condition search and threshold crossings, linear extrapolation, per-channel signal statistics, forward-kinematics consistency checks, and nominal-versus-actual absolute-difference comparison. The reference we want is what a competent engineer with normal instrumentation can extract from the telemetry, not what a person can do reading serialized numbers unaided. *No LLM assistance.* The annotator did not use a language model to produce or check any answer, so the baseline is independent of the systems under evaluation. *Identical items, identical scoring.* The annotator worked from the same items and prompts the model panel saw, and the answers were scored by the benchmark’s own grader: the same acceptance bounds for scalar and vector answers, the same exact-match rule for single-select and ranking, and the same per-position partial credit on multi-select. No separate human rubric was introduced, and the human receives no credit the panel would not receive. *Duration.* The entire process took a total of two work days of question answering and tool design by the expert.

Composition. 13 items at Level 1, 37 at Level 2, and 10 at Level 3, drawn across 14 distinct templates and covering all five answer formats. We plan on extending the baseline to address this current imbalance.

Results. Table 8 reports the expert’s score alongside the panel on the identical items. Note that these are *raw* scores, not the signed chance-corrected accuracies of Figure 3, and the two are therefore not directly comparable; the comparison of interest here is within the table, between the expert and the models on the same 60 items under the same grader.

The expert clears 0.96 at both Level 1 and Level 2 while the best model sits at 0.68 and the rest between 0.39 and 0.46. This settles the solvability question directly: the items are answerable from the released signals, so the low model scores reflect model limitations rather than broken or

Table 8: Human expert baseline against the model panel, raw scores on the identical 60-item sample. The expert reaches 0.917 overall (49/60 items exactly correct).

	Expert	Claude Sonnet 4.6	Mistral Large 3	GPT-5.1	Qwen3 235B	DeepSeek V3.2
Level 1	0.962	0.712	0.500	0.635	0.519	0.481
Level 2	0.980	0.662	0.444	0.392	0.385	0.351
Level 3	0.625	0.325	0.550	0.500	0.575	0.675

ambiguous items. It also shows that human experts are capable of a much deeper comprehension of machine state, whether healthy or anomalous, even though that comprehension is arrived at through a time-consuming process of measurement and computation (usually causing costly downtime) rather than intuitive signal comprehension.

The margin is not uniform across templates. It is narrowest where the answer can be read off a single channel with a short calculation, and widest on the templates that require committing to a specific root cause or a specific machine from the signal.

Level 3 is materially harder for the expert too. At Level 3 the expert reaches 0.625 with only 3 of 10 items exactly correct, and is not the top scorer on that level. This is generally explained by how undercovered and difficult the problem of counterfactual prediction is in time series analysis, leading to a lack of specific tools: the toolkit that carries the expert to near-ceiling on Levels 1–2 has no counterpart for reasoning about an intervention that was never executed. We read this as a property of the level rather than of the items, and it tempers the interpretation of the Level 3 column in Figure 3: the human ceiling there is much closer to the panel than at Levels 1–2.

Extension. We are extending the baseline to at least 100 Q&A pairs with a minimum of 25 per level, which means adding the Level 4 decision-making questions and broadening the Level 1 and Level 3 samples. We commit to reporting the full per-level results in an updated appendix for the camera-ready version, and to releasing the baseline itself: for every item, the expert’s answer, its score, and the method used to reach it.

F.3 Distribution-shift analysis

For the templates that produced a small noise-substitution Δ in the previous subsection, we confirm that they are nevertheless solvable by further empirical analysis, for example by inspecting how the sensor data distribution shifts when a fault is present. Figure 8 illustrates this by comparing multiple healthy KUKA episodes (blue) against multiple faulty episodes with an arm-disturbance fault (orange) drawn from FactoryWave (Figure 7 shows the physical setup of the fault for visual intuition), and reveals a clear distributional shift between the two groups on the channels the affected templates read from. The shift is more pronounced on some joints than on others, as the angle at which the disturbance is applied loads them unequally.

G Details of the FactoryWave dataset

FactoryWave was collected on two one-armed platforms: the **Universal Robots UR3** (cobot; OnRobot Screwdriver and two-finger gripper variants) and the **KUKA KR10** (industrial arm). Each episode corresponds to one complete task cycle; episodes are recorded under three conditions (normal, fault, and counterfactual), described below. Because state interpretation rather than policy quality is the object of study, motion is generated by a deterministic scripted controller. Every sample carries extensive labeling metadata, and the full native-rate sensor stream is logged without online feature selection. Table 9 gives the breakdown of the 8,983 episodes across robots, tasks, and conditions.

G.1 Tasks and phase structure

Three industrial tasks are represented in the dataset: Pick-and-Place, Screwing, and Peg-in-Hole. All three are executed on both robots, using matched scripted controllers and task layouts so that

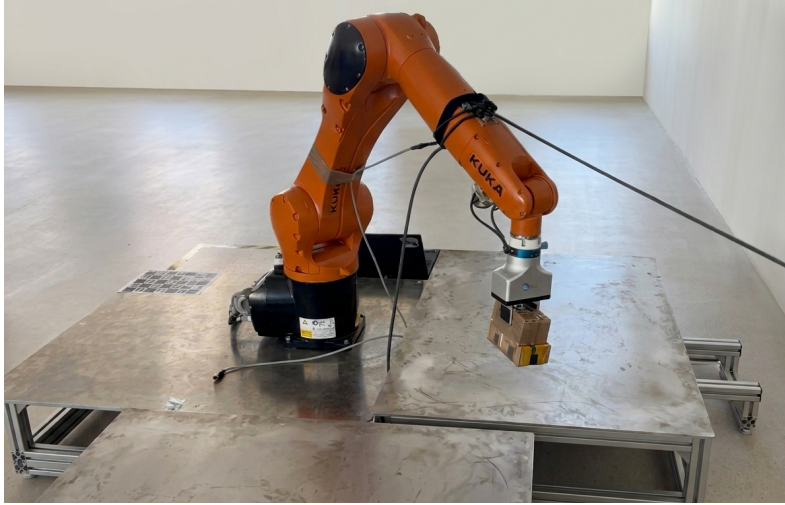


Figure 7: The arm-disturbance fault as physically applied to the KUKA in FactoryWave.

Table 9: Distribution of FactoryWave episodes by robot, task, and condition. Counterfactual entries count CF groups (one baseline plus one retained fault run per group), as each one represents data of one studied scenario.

Robot	Task	Normal	Fault	CF groups	Total
UR3	Pick-and-Place	1053	3238	109	4400
UR3	Screwing	230	850	40	1120
UR3	Peg-in-Hole	322	1200	100	1622
KUKA KR10	Pick-and-Place	578	1221	42	1841
Total		2183	6509	291	8983

cross-embodiment comparisons hold the task fixed. We structure each task as a deterministic sequence of phases (Tables 10, 11, 12); the phase index is written as a discrete label at every timestep, providing an interpretable segmentation that downstream models can condition on or that can be used to localize questions to a specific stage of the cycle. Within this fixed phase skeleton, we randomize as much of the episode as the task permits so that a question template that targets a given phase does not accidentally learn a fixed pose, timing, or trajectory. Pick and place locations are sampled uniformly from predefined workspace regions; phase-specific joint velocities, accelerations, and end-effector (EEF) rotations are drawn from bounded per-phase ranges; and approach angles to contact events are perturbed within the tolerances of the controller. This forces the Q&A pairs generated from FactoryWave to reflect phase-level reasoning rather than memorisation of a single scripted trajectory, which in turn makes them generalizable to a wide variety of trajectories and variation found in real recordings.

Table 10: Task phases for Pick-and-Place. EEF refers to End-Effector

Phase index	Phase name	Description
0	above_pick	EEF moves to a waypoint directly above the object
1	descend	EEF lowers toward the object
2	settle	Brief hold to let physics settle before grasping
3	close	Gripper closes around the object
4	lift	EEF lifts the grasped object off the surface
5	move_xy	Lateral transfer toward the bin
6	lower	EEF lowers into the bin
7	open	Gripper opens, object released
8	retract	EEF moves away from the bin
9	return	EEF returns to home configuration

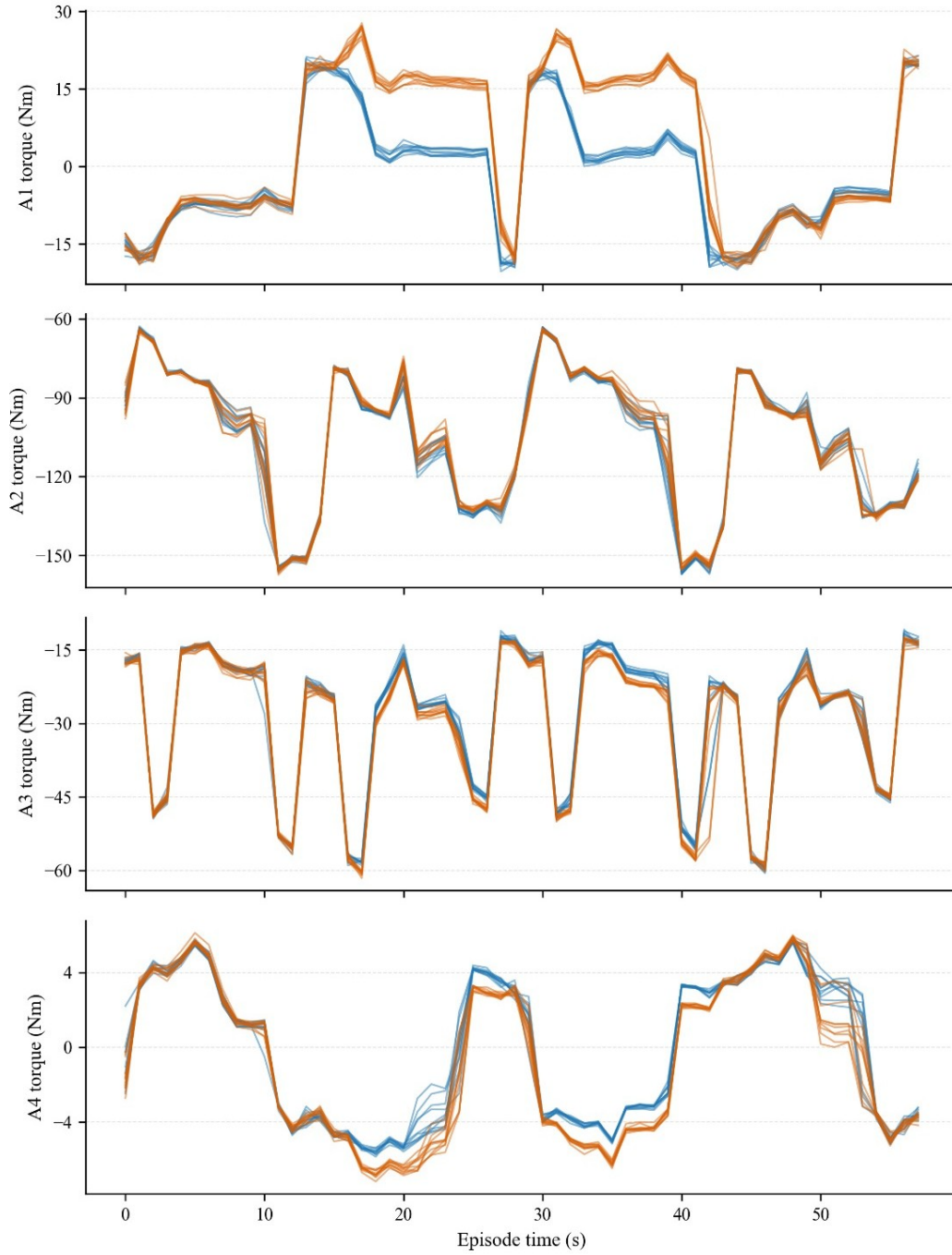


Figure 8: Sensor data distribution across multiple healthy KUKA episodes (blue) versus multiple faulty episodes with an arm-disturbance fault (orange) in FactoryWave. The visible shift between the two groups confirms that templates with small noise-substitution Δ are still solvable from the released signal; the magnitude of the shift varies across joints because the disturbance angle loads them unequally.

Table 11: Task phases for Screwing.

Phase index	Phase name	Description
0	approach	EEF moves to a pre-contact waypoint above the fastener
1	descend	EEF lowers to engage the fastener
2	screw	Continuous downward force + wrist rotation to thread
3	disengage	Gripper opens / tool lifts off
4	retract	EEF returns to a safe height
5	descend	EEF lowers to engage the fastener (loosening pass)
6	loosen	Loosen the screw
7	engage	Gripper closes on the retrieved screw
8	retract	EEF returns to home position

Table 12: Task phases for Peg-in-Hole.

Phase index	Phase name	Description
0	above_hole	EEF moves above the hole and aligns
1	insert	EEF pushes peg downward into the hole
2	disengage	Gripper opens
3	above_hole	EEF rises back to the waypoint above the hole
4	retract	EEF returns to home position
5	above_object	EEF moves above the object and aligns with it
6	engage	Gripper closes
7	above_object	EEF rises back to the waypoint above the object
8	retract	EEF returns to home position

1002 G.2 Episode metadata

1003 Alongside the sensor stream, every episode carries a fixed set of metadata fields describing the robot,
 1004 task, condition, payload, and end-effector configuration (Table 13). These fields are constant within
 1005 an episode and are used both as filters during question generation (e.g. to restrict a template to a
 1006 specific payload regime) and as provenance in the released Q&A pairs. This includes fault-specific
 1007 fields (such as injection timestamp or physical offset).

Table 13: Default per-episode metadata fields.

Field	Description
episode_id	Unique identifier for the episode
robot_model	Robot model (UR3, KUKA KR10)
task	Task (pick-and-place, screwing, peg-in-hole)
condition	Episode condition (normal, fault, counterfactual)
fault_id	Fault index for fault episodes; null otherwise
weight_of_box	Mass of the transported object in kg (pick-and-place only)
shape_of_box	Shape/dimensions of the transported object (pick-and-place only)
position_of_box	Initial XYZ position of the object on the table, where measurable
gripper_model	Gripper model and type (vacuum, two-finger, screwdriver)
payload_mass_configured	Payload mass configured at the controller (kg)
payload_cog_configured	Payload CoG offset configured at the controller (x, y, z in mm)
tcp_offset_configured	TCP position offset configured at the controller (x, y, z in mm)
tcp_orientation_configured	TCP orientation configured at the controller (rx, ry, rz in radians)

1008 G.3 Fault taxonomy

1009 FactoryWave includes 27 distinct fault types (Table 14). Each fault is paired with a plain-language
 1010 description of the underlying physical or control-level mechanism and the injection procedure used to
 1011 induce it during data collection. The last three columns indicate which tasks include the fault in the
 1012 collected dataset; faults that apply to all three tasks (e.g. collisions with a shared workspace object,
 1013 misconfigurations of the shared controller) are recorded under each applicable task with task-specific
 1014 signal imprints.

Table 14: Fault catalogue. Columns *PP*, *Scr*, *PiH* indicate whether the fault is included in the Pick-and-Place, Screwing, and Peg-in-Hole recordings.

Fault		Explanation	Injection	PP	Scr	PiH
Damaged thread	screw	Screw thread physically damaged, preventing proper engagement during tightening.	Pre-damaged the screw thread with sandpaper.	×	✓	×
Missing screw		Tightening is attempted with no screw present.	Removed the screw from screwdriver before the cycle.	×	✓	×
Damaged plate thread		The threaded hole in the plate is damaged, preventing engagement.	Pre-damaged the plate hole with a metal screwdriver.	×	✓	×
Loosening phase		A counterclockwise loosening rotation replaces tightening; a normal phase with a distinct signal. It is counted here as an anomaly to match the fault schema of the AURSAD dataset [1].	Reversed the rotation direction in the controller program to execute a counterclockwise loosening pass in place of tightening.	×	✓	×
Gripper activation failure		The vacuum gripper fails to activate and never picks up the box.	Disabled the gripper activation command in the robot program so the pick cycle completes without a grasp.	✓	×	×
Gripper release during motion		The gripper releases the payload mid-trajectory, causing an abrupt payload loss.	Triggered a programmatic gripper-release command at a scripted timestep mid-trajectory.	✓	×	×
Additional axis payload		A dead weight attached to one link increases inertia and gravity loading on all joints.	Bolted calibrated weights of varying mass to one robot link.	✓	✓	×
Collision with foam object		Contact with a soft foam block produces a brief TCP force spike without a protective stop.	Placed a foam cube directly in the programmed TCP trajectory.	✓	✓	✓
Unexpected payload weight		The transported box has a weight that deviates from the nominal payload configuration.	Swapped the nominal box for a known heavier or lighter one. This is done in a counterfactual context, ie. switching weights and asking about alternate weight scenario.	✓	×	×
Invalid gripping position		Timing error: the gripper closes after the arm has begun lifting, gripping the object off-center.	Inserted a brief delay in the controller program so gripper closure lags the lift motion.	✓	×	×
Unstable mounting platform	mounting	Base instability introduces low-frequency vibrations into the arm.	Placed 8 layers of towels under the base plate.	✓	×	×
Joint position limit violation		A joint moves outside its configured soft or hard position limits, triggering a safety stop.	Random waypoint slightly beyond the configured soft joint limit.	✓	×	×
TCP frame misconfiguration		TCP frame or mounting orientation misconfigured at the controller; gravity torques deviate.	Entered an incorrect TCP offset or mounting angle in the controller installation settings.	✓	✓	✓

Table 14 (cont.)

Fault	Explanation	Injection	PP	Scr	PiH
Payload weight mis-configuration	Payload mass misconfigured while a tool or workpiece is physically attached.	Left the configured payload weight at multiple wrong mass configurations while a task-dependent gripper remained mounted.	✓	✓	×
External arm disturbance	A continuous external force pulls or pushes the TCP during motion.	Anchored an elastic rope between the bench frame and the tool flange.	✓	✓	✓
Mild payload CoG misconfiguration	Payload CoG specified at an incorrect offset from the tool flange; gravity compensation is wrong.	Entered a CoG offset in the payload configuration that differs from the physical tool CoG.	✓	✓	✓
Collision with hanging cable	A loose cable drags along a robot link or the tool flange during motion.	Suspended a loose cable across the trajectory at arm height.	✓	✓	✓
Collision with cardboard object	A cardboard carton in the trajectory provides moderate resistance; may trigger a protective stop.	Placed a free-standing or braced cardboard box in the programmed trajectory.	✓	✓	✓
Collision with rigid object	A rigid plastic block in the TCP path triggers an immediate protective stop.	Clamped a rigid plastic block in the programmed TCP path. Unlike the other collisions, this fault consistently causes safety stops.	✓	✓	✓
Peg insertion misalignment	Peg approaches the hole at an angular or lateral offset and jams against the rim.	Added a recorded lateral offset at the insertion approach waypoint.	×	×	✓
Hole obstruction	Foreign object or debris in the hole prevents full insertion.	Placed a small object (metal screw) inside the hole before insertion.	×	×	✓
Incorrect insertion depth	Insertion terminates at an incorrect Z depth due to a misconfigured waypoint.	Shifted the insertion endpoint waypoint Z by recorded offset from nominal.	×	×	✓
Peg surface contamination	Contamination or roughness on the peg raises insertion friction and produces stick-slip.	Applied dry chalk powder to the peg surface.	×	×	✓
Fixture displacement	Insertion fixture shifted laterally, breaking the match between programmed approach and actual hole.	Physically shifted the hole fixture by recorded offset without updating the robot program.	×	×	✓
Self-collision	Arm configuration causes a link to contact the robot body or mounting fixture, triggering a protective stop.	Random waypoint that forces a self-colliding configuration or offset the mounting fixture into the trajectory.	✓	×	×
Missing box	Box absent from the pick position; the gripper closes on empty air.	Removed the box from the pick position before the episode started.	✓	×	×
Missing peg	Arm descends into the hole empty-handed; insertion produces no contact force.	Removed the peg from the gripper before the episode started.	×	×	✓

1015 G.4 Counterfactual episodes

1016 Counterfactual pairs are constructed for the subset of faults in Table 14 whose injection moment can be
 1017 controlled at a specific timestep rather than being present throughout the episode. For each such fault,
 1018 the initial conditions of the physical setup (object identity and pose, payload configuration, controller
 1019 settings) are held as constant as the platform allows, one baseline execution is recorded without any
 1020 injection, and several fault executions are then recorded under the same initial conditions with the
 1021 fault applied at a shared sampled timestep. Because the physical system is not perfectly reproducible,
 1022 the pre-injection sensor trajectories of the baseline and the fault candidates diverge slightly; we retain,
 1023 as the counterfactual partner of the baseline, the fault candidate whose pre-injection window is closest
 1024 to the baseline under the signature kernel MMD, and discard the others. This yields an approximate
 1025 counterfactual pair in which the early history is as matched as the real platform permits, while the
 1026 post-injection divergence isolates the effect of the intervention. The approximation error introduced
 1027 by this procedure, relative to the exact counterfactuals available in simulation, motivates the sim2real
 1028 analysis in Section K.

1029 G.5 Signal schema

1030 The UR3 is recorded at 125 Hz, yielding 136 per-sample signals grouped by semantic role (Table 15):
 1031 controller *setpoints*, actual *effort / feedback* readings from joints and the TCP, *context / health* signals
 1032 covering joint temperatures, voltages, and internal controller state, a small block of controller *status /*
 1033 *mode* indicators, and digital/analog *I/O* lines. A handful of episode-level provenance fields (episode
 1034 identifier, robot type, task, fault label, payload mass, recording date) are appended per episode but
 1035 are constant within an episode.

Table 15: UR3 signal groups recorded at 125 Hz (excluding episode-level provenance columns).

Group	# signals	Contents
Setpoint (intent)	42	Target joint position, velocity, acceleration, current, torque; target TCP pose and speed.
Effort / feedback	48	Actual joint position, velocity, current, current-as-torque, joint control output; actual TCP pose, speed, and force/torque.
Context / health	33	Joint temperatures, joint voltages, joint modes, tool accelerometer, raw F/T wrench, main/robot voltage and current, momentum, speed scaling, execution time.
Status / mode	7	Timestamp, robot mode, robot status, safety mode, safety status bits, runtime state, configured payload mass.
I/O	6	Digital inputs/outputs, two analog inputs, two analog outputs.
Total	136	

1036 The KUKA KR10 is controlled through the KRC4 controller. We stream 53 per-sample signals
 1037 at 83 Hz, grouped by semantic role in Table 16 using the same partition as the UR3 schema. A
 1038 handful of signals that the UR3 exposes natively are unavailable on KSS 8.3, specifically joint
 1039 velocities, commanded TCP pose, joint voltage, and joint-level control outputs; TCP force/torque
 1040 is also absent because no force sensor is fitted. To recover tool-side feedback, we mount a 9-DoF
 1041 inertial measurement unit on the gripper and log its triaxial linear acceleration (*acc_x/y/z*), angular
 1042 rate (*gyro_x/y/z*), and orientation (*angle_x/y/z*) at the controller’s 83 Hz cadence.

1043 H Time-series foundation model baseline (Chronos-Bolt)

1044 To check whether the LLM panel is the right comparison set on a benchmark whose motivation
 1045 references time-series models, we evaluate CHRONOS-BOLT [17] (~200 M parameters), a pretrained
 1046 probabilistic time-series forecaster, as a non-LLM baseline in the same zero-shot setting (no fine-
 1047 tuning, prompting, or context-window engineering on either side).

1048 **Template applicability.** A pure forecaster has neither a language interface nor a notion of options,
 1049 ranking, or counterfactual conditioning. Of the 21 active question templates in FactoryBench, only

Table 16: KUKA KR10 signal groups recorded at 83 Hz via RSI/KRL (excluding episode-level provenance columns).

Group	# signals	Contents
Setpoint (intent)	6	Commanded joint positions (degrees).
Effort / feedback	24	Actual joint positions (degrees), actual TCP pose (mm / degrees), per-axis motor current (% of max), and per-axis motor torque (Nm).
Context / health	11	Per-axis motor temperature (Kelvin), Cartesian acceleration on X, Y, Z and absolute magnitude (m/s^2), and speed override (%).
Tool IMU	9	Gripper-mounted 9-DoF IMU: triaxial linear acceleration acc_x/y/z (m/s^2), angular rate gyro_x/y/z (rad/s), and orientation angle_x/y/z (degrees).
Status / mode	1	Controller process state (FREE / ACTIVE / STOP / END).
I/O	2	Digital inputs bitmask and digital outputs bitmask.
Total	53	

three reduce to a well-posed forecasting problem: L1.7 (scalar forecast), L2.4 (scalar forecast under a known anomaly), and L2.5 (six-joint tensor forecast under a known anomaly). The remaining 18 templates (MCQ classification, ranking, counterfactual conditioning, free-form root-cause and protocol writing) are structurally outside the scope of any TSFM that lacks a language head. **This 14 % template coverage is itself a finding:** any TSFM, regardless of size or pretraining quality, can address only the forecasting slice of FactoryBench, and the other 86 % of the benchmark measures capabilities that no current TSFM provides.

Setup. For each forecasting item we form a univariate context per target channel, request the median quantile at the requested horizon, and score under the same chance-corrected piecewise-margin rule used in the main paper (Appendix A; $E = 1/4$). For the tensor template (L2.5) we run six independent per-joint forecasts and average the per-channel scores. Chronos-Bolt has no multivariate or text-conditioned mode, so cross-channel correlations and the anomaly named in the question are unavailable to it.

Results. On the three forecasting templates of the test split (Table 17, $n = 469$), Chronos-Bolt achieves chance-corrected accuracy of 51.5%, 46.9%, and 49.7% on L1.7, L2.4, and L2.5 respectively (signed correction). As a reference point, the strongest LLM in the panel reaches +46.8% on the L1 level aggregate (Section 5.1) and +25.8% on the L2 aggregate; a per-template comparison at that granularity is left to follow-up work.

Table 17: CHRONOS-BOLT zero-shot performance on the three forecasting templates of FactoryBench (test split, $n = 469$). Confidence intervals are 95 % percentile intervals from 1000 bootstrap resamples. Chance-corrected (CC) scores apply the signed transform $(s - 1/4)/(1 - 1/4)$.

Template	n	Raw mean	Raw 95% CI	CC mean	CC 95% CI
L1.7 (state, scalar)	334	0.636	[0.588, 0.681]	0.515	[0.451, 0.575]
L2.4 (intervention, scalar)	74	0.601	[0.493, 0.710]	0.469	[0.324, 0.614]
L2.5 (intervention, tensor[6])	61	0.623	[0.537, 0.708]	0.497	[0.383, 0.611]

Score distribution and signal coverage. Per-item scores are strongly bimodal: items tend to land at 1.0 or 0.0, with a smaller mass at the half-credit band, characteristic of a margin-thresholded scorer. Aggregating items by target-signal family shows roughly uniform capability across the four families (positions, speeds, torques/efforts, contact forces) within bootstrap noise, indicating the baseline does not selectively succeed on smooth signals and fail on noisy ones. Score declines monotonically with forecast horizon, as expected for a probabilistic forecaster.

Fairness of comparison. The comparison is apples-to-apples on item identity, scoring rule, chance correction, and zero-shot setting. It is *not* apples-to-apples on (i) input representation: LLMs see

1076 acronymized text-encoded values, alongside the question, robot description, and option set, while
 1077 Chronos-Bolt sees raw univariate float arrays only; and (ii) question awareness: on L2.4 and L2.5
 1078 the LLM is told which anomaly is present, while Chronos-Bolt cannot use that information. A
 1079 multivariate or text-conditioned TSFM (e.g. Moirai-MoE, Time-MoE) might exploit these signals;
 1080 we do not, to keep the baseline architecturally minimal. We therefore report Chronos-Bolt numbers
 1081 as template-specific rather than as level aggregates.

1082 **Interpretation.** Two messages emerge. **First, on the templates where forecasting is the right**
 1083 **tool, a 200 M-parameter pretrained forecaster is competitive with frontier LLMs**, consistent
 1084 with the main paper’s observation that LLMs struggle with precise numerical state extraction at L1.
 1085 The LLM panel is not the only, nor the most viable model class on the forecasting slice. This opens
 1086 the door to tool-assisted LLM agents that delegate forecasting subtasks to a specialised TSFM and
 1087 reserve the LLM for the linguistic, classification, and decision-making templates where it has the
 1088 structural advantage. **Second, the remaining 18 of 21 FactoryBench templates (classification,**
 1089 **ranking, counterfactual reasoning, free-form troubleshooting and optimization) are structurally**
 1090 **outside any current TSFM’s scope.** Together these reinforce the paper’s central claim: the gap
 1091 between current models and operational machine understanding is not closed by solely scaling LLMs
 1092 or deploying purpose-built forecasters.

1093 I Tool-augmented LLM agent baseline

1094 The zero-shot panel isolates the LLM’s ability to read industrial signals from text alone; the Chronos-
 1095 Bolt baseline isolates a specialised forecaster on the templates it can address. Neither captures the
 1096 design most likely to be deployed in practice: an LLM driver that delegates numerical subtasks to
 1097 purpose-built tools and reserves its own capacity for symbolic reasoning. We instantiate that design
 1098 as a ReAct-style [31] agent built on GPT-5.1 with four tools and evaluate it on the same benchmark.

1099 The composition is deliberately unambitious. Every component is off the shelf, the tools are the
 1100 obvious ones for the task, the driver is capped at six calls per item, and no part of the system is tuned
 1101 on FactoryBench. We built it this way on purpose: a strong bespoke agent would leave open whether
 1102 its gains came from tool use or from engineering effort invested in the agent itself, whereas gains
 1103 from a pipeline this plain are attributable to composition alone, and are a lower bound on what the
 1104 approach can deliver.

1105 **Pipeline.** The agent takes a FactoryBench item as input, decides which tools (if any) to invoke,
 1106 executes them, observes their outputs, and emits a final answer in the item’s expected format. It
 1107 has access to four tools: (i) a *signal-statistics* tool returning per-channel summaries (min, max,
 1108 mean, standard deviation, p_5 , p_{95} , derivative) computed from the item’s time-series context; (ii) a
 1109 *forecaster*, the same CHRONOS-BOLT (200 M) model used in Appendix H, called on a single channel
 1110 with a horizon the agent chooses; (iii) a *numerical sandbox*, a restricted Python interpreter with
 1111 standard scientific-computing libraries and the item’s time series pre-loaded; (iv) a *manual retriever*,
 1112 a dense-vector index over 9 vendor PDFs (UR3e, KUKA KR6/KR10 AGILUS, voraus-AD; 1,762
 1113 chunks embedded in 3072 dimensions), returning the three closest chunks for a natural-language
 1114 query. The driver LLM (GPT-5.1) invokes tools through a structured tool-calling interface, is capped
 1115 at 6 tool calls per item, and receives a system prompt that maps question shapes to preferred tools:
 1116 numerical prediction to the forecaster, segment localisation to the numerical sandbox, and root-cause
 1117 or remediation questions to the manual retriever.

1118 **Evaluation protocol.** We evaluate on a stratified random subset of the full test split: 200 items
 1119 per level (seed 42), sampled uniformly across templates within each level to preserve the paper’s
 1120 level-aggregate scoring rule. Every non-agent baseline in this appendix is re-evaluated on the same
 1121 200-item subset, so the per-level comparison is item-identical.

1122 **Results.** Table 18 reports chance-corrected accuracy on the 800-item evaluation subset. The
 1123 agent beats zero-shot GPT-5.1 at every level, with the largest gain on L1 (+22.9 pp), where the
 1124 two structurally hard slices (segment localisation on template L1.1 and short-horizon forecasting on
 1125 template L1.7) reward the numerical sandbox and the forecaster respectively. L2 also lifts substantially
 1126 (+13.8 pp) because tool-based statistics lookup helps on intervention templates that ask the LLM to

1127 compare two segments quantitatively. On L4 (decision-making, free-form judge-scored), retrieval of
 1128 vendor-specific remediation protocols lifts the raw judge score by +17.0 pp.

1129 Figure 9 shows the state loop and the per-level lift.

1130 **A modest pipeline recovers a large share of the gap.** We read these numbers as the encouraging
 1131 result of the paper. Nothing in the pipeline is novel, yet it more than doubles GPT-5.1 on two of four
 1132 levels and improves all four, which says the zero-shot scores understate what current models can do
 1133 on industrial telemetry once they are allowed to compute rather than only read. Two consequences
 1134 follow. First, a meaningful part of the failure the benchmark measures is a failure of *access*, not of
 1135 reasoning: the model can act on quantitative evidence when a tool puts that evidence within reach.
 1136 Second, and more useful to the community, the headroom above this baseline is large and cheap to
 1137 attack. The agent still trails the strongest zero-shot panel member on L1–L3 despite its tool access,
 1138 so the remaining gap is not merely a matter of wiring up more tools; and every choice we made here
 1139 (four tools, six calls, no tuning, no verification step, no multi-agent decomposition) is an axis a more
 1140 serious effort could push on. We therefore see FactoryBench less as a verdict on current models than
 1141 as a tractable target for agentic and tool-use research, where the first modest attempt already moves
 1142 the needle by double digits.

Table 18: Tool-augmented GPT-5.1 agent vs. GPT-5.1 zero-shot on FactoryBench, evaluated on a stratified random 200-item subset per level (seed 42). L1–L3 report signed chance-corrected accuracy ($\tilde{s} = (s - E)/(1 - E)$); L4 reports raw judge-ensemble accuracy under the L4 rubric (uncorrected, per Appendix A).

Level	GPT-5.1 zero-shot	GPT-5.1 agent (ours)	Δ
L1 (State, signed CC)	+15.9%	+38.8%	+22.9
L2 (Intervention, signed CC)	+1.3%	+15.2%	+13.8
L3 (Counterfactual, signed CC)	+18.2%	+24.9%	+6.6
L4 (Decision, raw judge)	16.2%	33.2%	+17.0

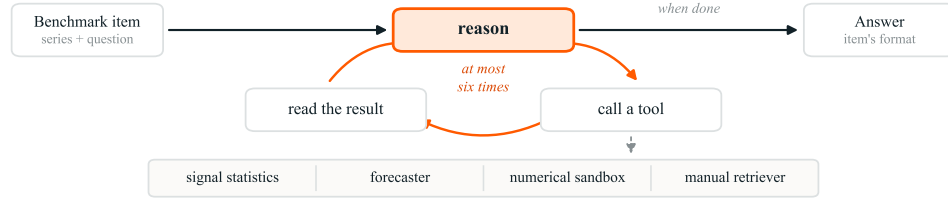
1143 **Tool usage.** Across the 800-item evaluation the agent issued $\sim 1,500$ tool calls, dominated by the
 1144 numerical sandbox (segment localisation and custom arithmetic), followed by the forecaster (T+N-
 1145 step value predictions), the signal-statistics tool (per-channel summaries), and the manual retriever
 1146 (fault-diagnosis and remediation queries at L4). The agent frequently answers identification-style
 1147 items without calling any tool, consistent with the system prompt’s directive to reserve tools for items
 1148 where the text prompt does not already contain the answer. Per-level tool distributions and average
 1149 call counts are released with the evaluation artefacts for reproducibility.

1150 **Interpretation.** The agent baseline confirms the paper’s core hypothesis in the direction predicted
 1151 by the Chronos-Bolt appendix: a specialised numerical tool is complementary to an LLM driver, and
 1152 the LLM+tools composition beats each part alone. The remaining gap to human-level performance
 1153 is largest on L3 (counterfactual reasoning, only +6.6 pp lift), where none of the four tools directly
 1154 address the reasoning demand; closing L3 is the natural next target for a domain-specific tool (e.g. a
 1155 causal-intervention simulator).

1156 J Example question-answer pairs and question templates

1157 Table 19 presents seventeen representative Q&A pairs spanning every causal level and every answer
 1158 format the generator produces (single-select MCQ, multi-select MCQ, tensor, ranking, free-form).
 1159 For readability we omit the underlying sensor context (tens to hundreds of time-series rows), truncate
 1160 verbose boilerplate such as “Answer only with . . .”, and abbreviate long option strings.

(a) The agent's state loop



(b) Four ordinary tools lift every level

L1–L3 signed chance-corrected · L4 uncorrected 0/0.5/1 rubric, not comparable across the L4 boundary · 800-item subset

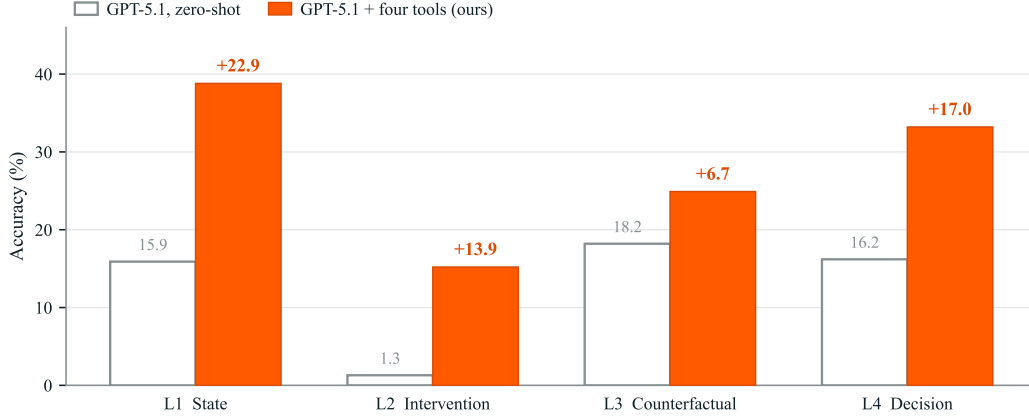


Figure 9: **A three-step loop, and the lift it produces.** (a) The agent as a state machine. The driver cycles through three states, reason, call a tool, read the result, and re-enters the cycle at most six times before leaving it to answer. Tools are reachable only from the calling state. Nothing is tuned on FactoryBench. (b) Accuracy per level for GPT-5.1 zero-shot and for the same model given those tools, on an 800-item stratified subset; L1 and L2 more than double. L1–L3 are signed chance-corrected; L4 uses the uncorrected free-form judge rubric and is not comparable across the L4 boundary.

Table 19: Representative Q&A pairs from FactoryBench spanning all four causal levels and every answer format.

Level	Answer format	Prompt, options, and reference answer
L1	Tensor	Q: The robot is performing a screwing task. We want to isolate the screwing phase in the robot’s time series. Assuming a fixed window length of 10 timesteps, at which timestamp should the window begin? A: 14 (accepted within ± 3)
L1	Single-select MCQ	Q: What changed between the two instances of robotic time series data? Options: a. different anomalous states (exactly one anomalous, or both but distinct) b. different tasks c. different robots d. same task, different phases A: a
L1	Single-select MCQ	Q: What robot does this sensor data originate from? Options: a. Agile Robots Yu 5 Industrial b. KUKA KR 10 R1100-2 c. Universal Robots UR3 A: c

continued on next page

Table 19 continued from previous page

Level	Answer format	Prompt, options, and reference answer
L1	Tensor	Q: Given the sensor stream below, what is the expected value of the position of joint 2 at T+606 ms? A: 102.7777
L2	Tensor	Q: The sensor stream below is from a robot exhibiting a collision with a cardboard object. What is the expected value of the velocity of joint 3 at T+68 ms? A: -0.272044 (accepted within ± 2.42)
L2	Tensor	Q: The sensor stream below is from a robot exhibiting a collision with a cardboard object. What are the expected values of joint positions at T+506 ms? A: [17.884, -115.897, 121.049, -95.595, -90.258, 293.860]
L2	Tensor	Q: Knowing that the robot suffers from a hole obstruction, we want to isolate the “rise back above the object” phase. Assuming a fixed window length of 16 timesteps, at which timestamp should the window begin? A: 202 (accepted within ± 3)
L2	Single-select MCQ	Q: Given the sensor data, determine what anomaly is present. Options: a. sustained command/measurement deviation b. no anomaly c. measured joint speeds drop abruptly with a TCP force spike (typical collision signature) d. isolated unrealistic force-sensor spike inconsistent with motion A: c
L3	Ranking	Q: Given the baseline stream and the counterfactual scenario where a collision foam object occurs at timestep 7366 ms, rank the four labeled signal segments in the order they would appear as the event manifests. A: abdc
L3	Multi-select MCQ	Q: Given the counterfactual scenario where a collision hanging cable occurs at timestep 8363 ms, select all statements that would apply. Options: a. motor current rises $\geq 33\%$ while speed stays $\leq 10\%$ of baseline (stall-like) b. contact-force spike $\geq 36\%$ above baseline c. tracking-error rise $\geq 34\%$ above pre-event mean d. at least one joint speed drops sharply ($\geq 31\%$) A: FFFT
L3	Tensor	Q: In the counterfactual scenario where a collision foam object occurs at timestep 7659 ms, what would be the expected value of the position of joint 2 at T+100 ms? A: 72.144939 (accepted within ± 2.77)
L4	Free form	Q: Given the sensor stream below, does the machine show signs of anomalous behavior? If yes, identify the most likely root cause and describe the steps you would take to fix it. A: No anomalous behavior detected; the machine is operating normally; no remediation is required.
L4	Free form	Q: Given the sensor stream below, does the machine show signs of anomalous behavior? If yes, identify the most likely root cause and describe the steps you would take to fix it. A: a. Inspect the workspace and remove any soft obstructions from the programmed trajectory. b. If a protective stop occurred, acknowledge it in the robot controller and verify arm posture before resuming. c. Reduce speed if repeated false-positive collision detections occur.

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Table 19 continued from previous page

Level	Answer format	Prompt, options, and reference answer
L4	Free form	Q: Given the sensor stream below, does the machine show signs of anomalous behavior? If yes, identify the most likely root cause and describe the steps you would take to fix it. A: a. Stop the robot and update the payload configuration to include the additional axis weight. b. Set the correct CoG offset for the modified arm in the installation settings. c. Reduce motion accelerations and speeds to account for increased effective inertia. d. Run a slow-speed test cycle to verify dynamic loads stay within rated limits before resuming.
L4	Free form	Q: An engineer wants to increase the effectiveness and accuracy of this machine. Based on the sensor stream below, what operational changes or parameter adjustments could help achieve this? A: The payload mass is set to 0.0 kg but the actual payload weighs 1.5 kg. Update the payload mass in the installation settings to 1.5 kg.
L4	Free form	Q: An engineer wants to increase the effectiveness and accuracy of this machine. Based on the sensor stream below, what operational changes or parameter adjustments could help achieve this? A: The TCP offset is misconfigured at [0, 0.3, 55] instead of the correct [0, 0, 55]. Update the TCP position offset in the installation settings to [0, 0, 55].
L4	Free form	Q: An engineer wants to increase the effectiveness and accuracy of this machine. Based on the sensor stream below, what operational changes or parameter adjustments could help achieve this? A: The payload center of gravity is set to [25, 0, 20] but the correct value is [0, 0, 20]. Update the CoG offset in the installation settings to [0, 0, 20].

1161 K Simulation setup and sim-to-real gap analysis

1162 We run sim2real rollouts in Isaac Sim using the built-in UR3 asset, with phase-consistent waypoint re-
 1163 play of the recorded `target_joint_*` signals. The simulated task is UR3 pick-and-place, following
 1164 the 10-phase structure of Table 10. Payload settings are matched episode-wise to the FactoryWave
 1165 chronology.

1166 Each real episode is exported, converted to a per-episode simulation config, and replayed in Isaac.
 1167 Simulated outputs preserve the FactoryWave field schema (`joint_*`, `tcp_*`, `task_phase`, etc.)
 1168 and are paired with their real counterpart by episode ID. Gap metrics are computed phase-wise on
 1169 aligned trajectories: joint RMSE (deg), TCP position RMSE (mm), TCP rotation error (mrad), and
 1170 Wasserstein-1 on `joint_current_*` (A).

1171 Across 1,155 paired episodes, pooled per-episode metrics are: joint RMSE mean/median =
 1172 3.65/2.83 deg, TCP position RMSE = 13.16/13.26 mm, and W1 effort mean = 0.83/0.81 A. TCP
 1173 rotation error remains broad (median 2605 mrad), indicating orientation mismatch is the least stable
 1174 component. Translational kinematics and effort-proxy alignment are substantially more consistent
 1175 than rotational alignment.

1176 L Representation probing: does the model encode what it cannot report?

1177 FactoryBench, as presented in the main text, is a purely *behavioural* evaluation: a model is prompted
 1178 with telemetry and graded on its final answer. A wrong answer therefore conflates two qualitatively
 1179 different failure modes: (a) the relevant machine concept is *absent* from the model’s internal repre-
 1180 sentation, or (b) the concept is *present* in the representation but the model cannot surface it through
 1181 its output. Distinguishing them matters for interpreting the Level-4 collapse of Section 5.3, and for
 1182 localising where progress must come from (better decoding versus better pretraining). We separate
 1183 the two with a targeted linear-probing study [54] on an open-weight panel member, in the spirit of the

1184 geometric analyses of motion transformers in *Words in Motion* [55]. All numbers below reproduce
1185 from the released probing code.

1186 **Setup.** We probe **Qwen3-4B** (frozen, no fine-tuning). Each item is passed through the model once
1187 on the *exact* prompt the zero-shot panel saw, and we cache the residual-stream activation at every
1188 layer (0=embedding through 36) at two read-out positions: the last prompt token, and the mean over
1189 the time-series token span. All activations are computed with causal attention, verified by a startup
1190 self-test that checks that layer- k predictions are invariant to removing later tokens.

1191 **Probes and controls.** For each (concept, layer, read-out) we fit an L_2 logistic-regression probe on
1192 the top-256 principal components of the standardised activations, with the standardiser and PCA fit
1193 *inside* each cross-validation fold so no validation point shapes its own projection. The regularisation
1194 strength is chosen by *episode-grouped* 3-fold cross-validation, and the reported layer is the one
1195 that maximises this CV score (never the test accuracy, to avoid a layer-selection peek). A one-
1196 hidden-layer MLP is a non-linear reference. Four controls guard the read-out: (i) **selectivity** [56],
1197 the probe accuracy minus that of an identical probe trained on *shuffled* labels, which stays near zero
1198 unless the probe reads a real concept rather than memorising; (ii) a **raw-signal baseline**, the same
1199 probe fit on per-channel signal statistics (mean/std/min/max/last, with the timestamp stripped so no
1200 positional information leaks), which measures what is decodable from the numbers with *no model*;
1201 (iii) a **random-init control**, the same probe pipeline on a same-architecture network with *untrained*
1202 weights, which measures how much of the decodability is generic random-feature structure of the
1203 input rather than something the model learned in pretraining; and (iv) **episode-disjoint** train/test
1204 splits with bootstrap 95% confidence intervals that resample whole *episodes* (not items, which are
1205 correlated within an episode). Headline probe accuracies are the mean over three random seeds (split,
1206 PCA, and C -selection). We probe two machine concepts, plus a *source-dataset* positive control and a
1207 *task-phase* leakage check. Neither concept is a benchmark label in its own right: both are derived
1208 from the injected `fault_id` recorded with each episode. *Fault type* collapses that identifier to the six
1209 most frequent injected mechanisms, chosen so the classes carry comparable numbers of items and a
1210 six-way probe is not dominated by one mechanism; the long tail of rarer faults is excluded rather than
1211 merged into a catch-all, which would make the sixth class a mixture and the accuracy uninterpretable.
1212 *Fault presence* is the binary reduction of the same identifier, healthy against any injected fault.

1213 **Behavioural read-out a .** To place each probe against the model’s own competence we ask the
1214 same frozen model the concept directly, on the same held-out probe items: fault presence as a yes/no
1215 question, and fault type as an A–F multiple choice over exactly the probe’s six classes, with the
1216 option-to-letter assignment *randomised per item* so a positional answer bias cannot masquerade as
1217 knowledge (scored against the gold `fault_id`). We let the model reason (chain-of-thought, up to
1218 1024 tokens for the six-way question), take its committed answer when it gives one, and for items
1219 where it reasons without committing we read its choice *parse-free* from the logits over the option
1220 tokens after a “therefore the answer is” cue — so every item is scored and no inference is discarded.
1221 For fault presence we add a **negated-wording control**: we re-ask “is this episode healthy?” and
1222 check whether the model’s answer flips, to separate a genuine reading of the signal from a fixed
1223 yes-token bias. Probe accuracy p and behavioural accuracy a are computed on the identical test items,
1224 giving a matched read: p asks whether the concept is decodable, a whether the model uses it.

1225 **Fault presence is decoded but not reliably reported.** Fault presence is linearly decodable at
1226 $p=0.83$ (three-seed mean 0.86 ± 0.02 ; mean-pool 0.86; selectivity 0.34, so this is not probe memori-
1227 sation). The model’s behavioural read-out, by contrast, is unreliable and *framing-sensitive*. On the
1228 benchmark’s own anomaly wording it answers “yes” to *every* episode and scores $a=0.37$ — below
1229 the 0.5 chance of random guessing, because only 38% of episodes are faulty, so an always-“anomaly”
1230 policy scores at that base rate (Figure 10, left). Re-worded to “is this episode healthy?” it is less de-
1231 generate (it answers “no” on 64% of episodes) but still reaches only $a_{\text{neg}}=0.54$, barely above chance.
1232 So the model has no fixed yes-token reflex — its answer flips with wording — but a miscalibrated,
1233 framing-sensitive bias toward predicting a fault; under *neither* framing does it approach the 0.83
1234 that is linearly available, so the read-out gap is robust to the elicitation rather than an artefact of the
1235 harsher wording. The gap is moreover entirely at the read-out: the raw-signal baseline (0.83) and
1236 the random-init network (0.82) both decode fault presence just as well, so it is an “easy” property
1237 already carried by signal statistics and even by untrained features, to which the pretrained model adds
1238 no representational value.

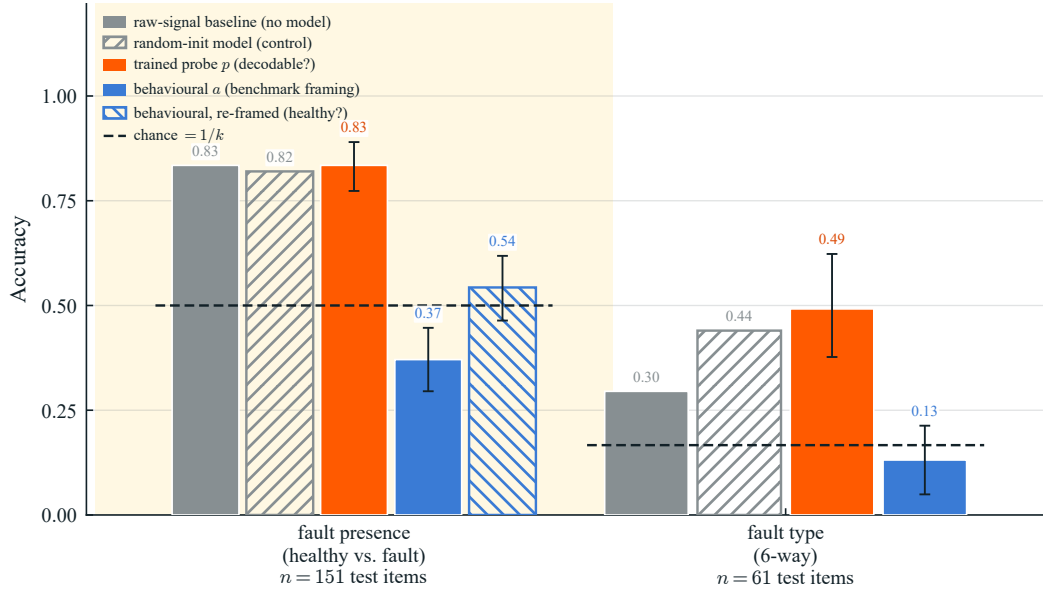


Figure 10: **Decodable is not the same as learned, and neither is the same as reported.** For fault presence and fault type we compare the trained linear probe’s accuracy p (is the concept decodable from Qwen3-4B’s frozen residual stream?) with the model’s own behavioural accuracy a on the same question, alongside a raw-signal baseline (no model), a random-init control (same architecture, untrained weights), and chance. Fault presence is decodable at $p=0.83$, but the model’s read-out is framing-sensitive and never approaches it: $a=0.37$ under the benchmark’s anomaly wording (it replies “anomalous: yes” to every episode) and $a_{\text{neg}}=0.54$ when re-worded to “is this episode healthy?” — both near the 0.5 chance and far below 0.83. Raw signal statistics (0.83) and even an untrained network (0.82) decode it just as well, so this availability is surface, not learned. Fault type is only weakly decodable ($p=0.49$, CV-selected) and, crucially, *no better than the random-init control* (0.44); the model answers at $a=0.13$. Chance is $1/k$ (uniform random guessing): 0.5 for the binary presence task, $1/6$ for the six-way type task. Error bars are bootstrap 95% CIs over episodes; accuracies are on the held-out test split.

1239 **Fault type is neither learned nor reported.** Fault *type* is only weakly decodable: $p=0.45\pm0.07$
1240 (three-seed mean; the CV-selected layer is shallow, e.g. layer 1), above the raw-signal baseline (0.30)
1241 but *not* above the random-init control (0.44). All three clear the $1/6\approx0.17$ chance of the six-way
1242 task, so the signal does carry some fault-type structure — but the trained probe extracts no more of it
1243 than an untrained network. A same-architecture network with untrained weights decodes fault type as
1244 well as the pretrained one, and its per-layer curve tracks the trained probe at every depth (Figure 11a);
1245 the trained model’s marginal contribution over an untrained one is within noise. This decodability
1246 therefore reflects generic random-feature structure of the input rather than a representation the model
1247 learned in pretraining. (An unregularised argmax over all layers and read-outs reaches 0.57 at layer 31
1248 on the test set, but that is a selection artefact: the CV-selected layer, which never sees the test set, is
1249 shallow and near 0.45.) Behaviourally the model reaches only $a=0.13$, below the $1/6$ chance of the
1250 six-way task. Given a chain-of-thought budget it does commit an answer on 84% of items, and its
1251 errors now spread across five of the six classes — the earlier apparent “collapse onto two classes”
1252 was an artefact of a fixed option order, which we remove by randomising the A–F assignment per
1253 item. For fault type, then, we find neither a learned representation nor a usable read-out.

1254 **Positive control and leakage check.** The source-dataset positive control is recovered at 1.00
1255 (selectivity 0.83; raw-signal baseline 1.00), confirming that the rig surfaces a genuinely present
1256 concept and that the weaker results elsewhere are real, not a broken probe. The task-phase concept
1257 probes at a perfect 1.00 from the last token but only 0.59 from the time-series tokens (Figure 11b):
1258 because the phase name is written into the Level-1 question wording, this is lexical availability, not
1259 time-series understanding, so we exclude it from the claims.

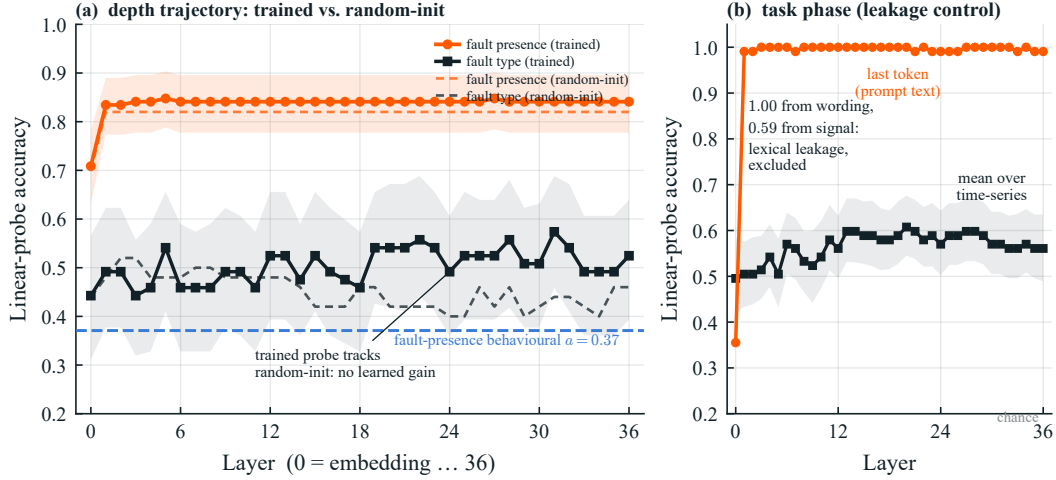


Figure 11: **An untrained network decodes fault type as well as the trained one.** (a) Per-layer linear-probe accuracy (last-token read-out), trained (solid) vs. random-init control (dashed). Fault presence is decodable already at the embedding (0.71) and saturates immediately, but the random-init curve matches it. Fault type’s trained curve tracks its random-init curve at *every* depth — no learned gain — and the CV-selected best is shallow, not the test-set argmax at L31. The model’s behavioural accuracy on fault presence ($a=0.37$, dashed blue) lies below the probe at every depth, so the read-out gap is not an artefact of layer choice. Bands are episode-level bootstrap 95% CIs. (b) Leakage control: the task-phase concept probes to 1.0 from the last prompt token (the phase name is written in the question) but only ~ 0.59 from the mean-pooled time-series tokens, so we exclude it.

Takeaway. The two concepts sit at different points on a representation–read-out spectrum. Fault *presence* is a *read-out* failure: it is trivially decodable ($p=0.83$, matched by raw signal statistics and by an untrained network), yet the model answers near chance under either framing (0.37–0.54), applying an anomaly-predicting policy it cannot calibrate. Fault *type* is weaker still: the pretrained model shows no decodable representation beyond a random-init baseline (0.45 vs. 0.44) *and* cannot report it ($a=0.13$) — a representational gap, not merely a decoding one. Part of the Level-4 collapse of Section 5.3 is therefore a genuine read-out/calibration failure where the concept is present, and part reflects the absence of a learned representation where it is not. Near-term progress will need both: better decoding and calibration for the surface concepts, and better representation learning — from pretraining or fine-tuning on a frozen backbone — for the computed ones.

Scope. The study probes a single open-weight model (the only panel member exposing activations and fitting the long prompts) and two concepts with dense ground truth. Two cautions on interpretation. First, a linear probe measures a *lower bound* on decodability under a specific read-out: a near-chance probe bounds what a linear read-out recovers, not the information content of the signal, so our fault-type result is “the model does not linearly represent this beyond random features,” not “the information is absent” (the random-init control already recovers 0.44, so the signal is not empty). Second, the probe is supervised on frozen features whereas the model answers zero-shot, an asymmetry intrinsic to probing, where p measures latent *availability* and a measures zero-shot *usability*. Full per-layer numbers for both read-outs, all four controls, the multi-seed spreads, and the exact reproduction command are released with the benchmark code.

M Causal schema (SCE): full details

This appendix gives the long-form definition of the Setpoint–Context–Effort + Feedback (SCE) schema used to organize every time-series channel in FactoryBench, together with the residual-based fault definition that the schema enables. The motivation is summarized in Section 3.2; here we cover the per-group channel breakdown, the residual formalism, and the per-robot calibration that makes faults computable rather than imputed.

1286 The schema groups every recorded channel into one of three causal roles. **Setpoint** captures the
 1287 controller’s commanded target: per-axis position, velocity, and acceleration setpoints emitted by the
 1288 trajectory generator. **Context** captures physical and configured conditions affecting behavior, split
 1289 into *static* context (episode-level metadata such as payload mass, payload center of gravity, TCP
 1290 offset, gripper model, and box / peg geometry) and *dynamic* context (per-sample channels such as
 1291 joint temperatures, supply voltages, controller mode, and safety mode). **Effort + Feedback** captures
 1292 the machine’s measured response: motor currents and torques (effort), measured joint positions and
 1293 TCP pose (feedback), tool accelerometer, and acoustic emission where available. The exact per-robot
 1294 channel mapping appears in the signal-group tables in this appendix (UR3 and KUKA KR10).

This split enables a single operational definition of a fault that applies across machines and tasks. Under healthy operation, the measured Effort + Feedback response y_t is by definition a function of the Setpoint command S and the Context C_t :

$$y_t = f(S, C_t),$$

1295 where f represents the nominal plant dynamics (inverse-dynamics-plus-controller transfer) calibrated
 1296 per robot. Faults manifest as structured residuals from this expected baseline: $y_t - f(S, C_t)$ forms
 1297 a clear, fault-specific spatiotemporal pattern. Because every episode supplies the paired (S, C_t, y_t)
 1298 streams, this residual is computable directly from the data, giving ground truth that does not have to
 1299 be hand-imputed. Figure 12 summarises the causal relationships between the three groups.

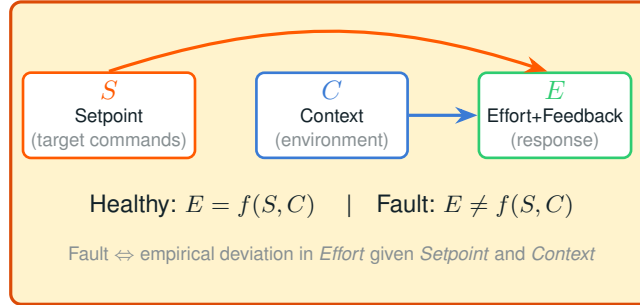


Figure 12: Physical causal schema. Setpoint commands S and contextual variables C jointly determine the measured Effort + Feedback response E under healthy operation. Faults manifest as residuals $E - f(S, C)$ from this nominal mapping.

1300 N License and availability

1301 FactoryBench and FactoryWave are freely available for academic and commercial use under a two-
 1302 track licence: the **generator source code, evaluation scripts, and tooling** (this repository) are
 1303 released under the **MIT License**, and the **data and benchmark artefacts** (FactoryWave episodes,
 1304 question templates, paraphrase banks, LLM-as-judge prompts, and the full Q&A dataset) are re-
 1305 leased under **Creative Commons Attribution 4.0 (CC BY 4.0)**, with attribution to be given to the
 1306 FactoryBench authors. Both artefact tracks are distributed via the public Hugging Face repository
 1307 FactoryBench/FactoryBench. To guard against test-set contamination in future model releases,
 1308 we hold back a private 15% subset of the Q&A pairs at the *episode* level (so all questions grounded in
 1309 any held-out episode stay private across all four levels) and keep its raw episode data, gold answers,
 1310 and rubric annotations off the public release; this private slice is used internally for contamination
 1311 checks and may also be used to score future evaluatees on request. The remaining 85% is fully public,
 1312 with a train/validation/test split included in the release so that evaluatees can reproduce the exact
 1313 partition used in this paper.

1314 We provide a versioned release track (e.g., v1.0, v1.1) so that reported numbers always refer
 1315 to a fixed benchmark state; corrections to labels or templates are published as minor-version
 1316 updates with a public changelog, and the exact version used in any evaluation is stamped into
 1317 every result file. **All counts, tables, and results reported in this paper correspond to re-**
 1318 **lease FactoryBench/FactoryBench:v1.0.0, the paper-submission snapshot** (70,915 released
 1319 Q&A items grounded in 8,983 FactoryWave episodes). The live main branch of the dataset

repository may contain additional episodes and Q&A items added after the submission snapshot; those additions are versioned in the changelog and do not retroactively affect the numbers here. Bug reports and label-correction requests are tracked on the public repository at <https://github.com/Forgis-Labs/Factorybench>.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and the Introduction section together introduce FactoryBench as a four-level benchmark grounded in FactoryWave (our new dataset) plus AURSAD [1] and voraus-AD [2], with an LLM-as-judge protocol for Level 4 free-form items; every claim is backed by the results in the Evaluation section and the dataset, scoring and template details in the appendices.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The Limitations subsection (under Evaluation) explicitly discusses the two main limitations: the approximate nature of physical Level-3 counterfactual ground truth (signature-kernel-MMD pair selection rather than exact do-operations), and the simplification of faults to atomic, well-isolated mechanisms drawn from a closed catalogue of 27 injected faults.

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- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
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- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.

- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is a benchmark/dataset contribution and does not include theoretical results requiring formal proofs.

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- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The model panel and decoding settings (Section 5.1), the per-format scoring rules and chance-correction formulas plus verbatim Level-4 judge prompts (Appendix A), the per-template feature lists and question templates (Appendix B.1), the FactoryWave recording and signal schema (Appendix G), and the released generator code together specify how to reproduce every reported number.

Guidelines:

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- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
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- (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: FactoryBench, FactoryWave, the question templates, paraphrase banks, LLM-as-judge prompts, generator source code, and the full Q&A dataset are released under the MIT License via the public Hugging Face repository FactoryBench/FactoryBench, alongside a versioned GitHub repository; see Appendix N for the license, versioning, and release-track details.

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Justification: The zero-shot evaluation panel and prompting protocol are described in Section 5.1; per-format scoring rules, chance-correction, and the LLM-as-judge prompts and aggregation are in Appendix A; per-template feature pre-selection and the 10 Hz resampling pipeline are in Appendix E.

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7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Each model is evaluated in a single zero-shot pass at the provider's default decoding settings, so we do not report per-item error bars. Judge variance on Level-4 is mitigated by aggregating three independent LLM judges (GPT-5.1, Claude Sonnet 4.6, DeepSeek V3.2) by median rather than by repeated sampling.

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Justification: Q&A generation runs on a single modern laptop and is not compute-heavy. Closed-weight models (GPT-5.1, Claude Sonnet 4.6, Mistral Large 3) are evaluated via their providers' APIs; open-weight models (DeepSeek V3.2, Qwen3-235B, Qwen3-4B) through hosted inference. Total API cost for the full evaluation across all six models was approximately \$350.

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1537 available APIs at submission time.

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1550 tive maintenance, and engineer-assistant deployments on real factory robotic systems) is
1551 described in the Introduction. Direct negative impacts are limited because the released data
1552 are sensor-only with no human subjects, and the benchmark is explicitly designed to surface
1553 failure modes of current LLMs at Level 4 rather than to push deployment of unverified
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Answer: [Yes]

Justification: LLMs are central to the Level-4 free-form scoring methodology: a three-judge ensemble (GPT-5.1, Claude Sonnet 4.6, DeepSeek V3.2) scores every L4 reply against the reference under template-specific rubrics, with the per-item score taken as the median of the three votes. The verbatim troubleshooting and optimization judge prompts and the aggregation protocol are documented in Appendix A.

1687

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